

Bihar Electricity Grid Code, 2010

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State of Bihar - Act

Bihar Electricity Grid Code, 2010

BIHAR

India

Bihar Electricity Grid Code, 2010

Rule BIHAR-ELECTRICITY-GRID-CODE-2010 of 2010

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Bihar Electricity Grid Code, 2010

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Notification No. BERC/Regl-1/2009 02-238. - In exercise of powers conferred under Section 181 read with Clause (h) sub-section (1) of Section 86 of the Electricity Act, 2003 and all powers enabling it on that behalf, the Bihar Electricity Regulatory Commission hereby specifies the following Grid Code to be known as Bihar Electricity Grid Code, 2010 which shall come into force from the date of notification. This Grid Code is applicable for the grids in the State of Bihar only.

1.

Introduction.

- 1.1. (1) The Grid Code lays down the Rules, Guidelines and Standards to be followed by all Users of Grid in the State

of Bihar to plan, develop, operate and maintain the power system in the State in the most efficient, reliable, economic and secure manner in integration with the Eastern Regional Grid as per the provisions of Indian Electricity Grid Code (IEGC) notified by the Central Electricity Regulatory Commission (CERC) from time to time.

(2)

In terms of Section 39(1) of the Electricity Act 2003, the State Government is required to notify the Board or a Government Company as State Transmission Utility (STU). State Government has notified the Bihar State Electricity Board (BSEB) as the State Transmission Utility (STU) which is a deemed licensee at present in terms of Section 172 of the Electricity Act, 2003.

(3)

As per Section 39(2) of the Electricity Act, 2003 the functions of State Transmission Utility are as follows

(a)

to undertake transmission of energy through the Intra-State Transmission System

(b)

to discharge all functions of planning and coordination relating to Intra-State Transmission System with

(i)

Central Transmission Utility;

(ii)

State Governments;

(iii)

Generating Companies;

(iv)

Regional Power Committee;

(v)

Authority;

(vi)

Licensees;

(vii)

Any other person notified by the State Government in this behalf

(c)

to ensure development of an efficient, coordinated and economical system of intra-state transmission lines for smooth flow of electricity from a generating station to load centres;

(d)

to provide non-discriminatory open access to its transmission system for use by -

(i)

any licensee or generating company on payment of the transmission charges;

(ii)

any consumer as and when such open access is provided by the State Commission under sub-section (2) of Section 42, on payment of the transmission charges and a surcharge thereon, as may be specified by the State Commission.

(4)

STU shall not engage in the business of trading of electricity.

(5)

As per the Section 31 of Electricity Act, 2003, the State Government shall establish a centre to be known as the State Load Despatch Centre (SLDC) and the State Load Despatch Centre shall be operated by a Government company or any authority or corporation established or constituted by or under any State Act, as may be notified by the State Government. Until a Government Company or any Authority or Corporation is notified by the State Government, the State Transmission Utility shall operate the State Load Despatch Centre, which shall:

(a)

be responsible for optimum scheduling and despatch of electricity within a State, in accordance with the contracts entered into with the licensees or the generating companies operating in that State;

(b)

monitor the grid operations;

- (c)
keep accounts for the quantity of electricity transmitted through the State Grid;
- (d)
exercise supervision and control over the Intra State Transmission system;
- (e)
be responsible for carrying out real time operations for grid control and despatch of electricity within the State through secure and economic operation of the State Grid in accordance with the Grid Standards and the State Grid Code.
- (6)
The SLDC and licensees shall comply with; the directions of Regional Load Despatch Centre in connection with integrated grid operation of the power system or in regard to matter which may affect the operation of the Inter State Transmission System.
- (7)
SLDC shall not engage in the business of trading of electricity.
- 1.2
Objective. - The Grid Code is designed to facilitate the development, operation maintenance and planning of an efficient, coordinated and economical power grid by specifying principles and procedures for STU and all the Users connected to that system. It seeks to be non-discriminatory and ensures that interfaces are not areas of weakness in the supply chain.
- 1.3
Structure of the Grid Code. - The Grid Code comprises of following chapters:
 - I. General Code. - General Code includes Sections on:
 - (a)
Management intended to ensure that all other Sections of the Grid Code work together in the management of the Grid Code
 - (b)
Review Principles: specify a procedure for review of Grid Code to cater to inadvertent omissions and any modifications needed from time to time.
 - II. Planning Code. - Planning Code includes Sections on:
 - (a)
System Planning: specifies the procedures to be applied by STU in the planning and development of the State Transmission System and by other Users connected or seeking Connection to the State Transmission System.
 - (b)
Principles: specify procedures to be followed by STU in the development of the State Transmission System in the long term taking into account the requirements for new connection of generation and demand.
 - (c)
Connection conditions: specifies the technical requirements and standards to be complied with by STU and other Users connected or seeking connection to the State Transmission System.
 - III. System Operation Code. - System Operation Code includes Sections on:
 - (a)
System Operation: Specifies the conditions under which STU shall operate the State Transmission System, the Generating Companies shall operate their Power Stations and the Distribution Licensees shall operate their Distribution Systems in so far as necessary to ensure the security and quality of supply and safe operation of the State Transmission System under both normal and abnormal operating conditions.
 - (b)
Outage Planning: Specifies the procedures relating to the coordination of outages for scheduled maintenance of the State Transmission System, Generating Units, embedded within the State system and not classified as regional utilities, Captive Power Plant (CPP) and Distribution System that will use the State Transmission System.
The Generating stations based on non-conventional/renewable energy sources with generating power capacity of above 10 MW will also come under this purview.
 - IV. Schedule and Despatch Code. - Schedule and Despatch. - Specifies the procedures relating to the scheduling and despatch of Generating Units and drawal by Distribution Licensees / Open Access Customers to meet State demand and drawal allocation.

V. Protection Code. - Protection Code specifies the co-ordination responsibility and minimum standards of protection that are required to be installed by Users of the State Transmission System.

VI. Metering Code. - Metering Code specifies the minimum operational and commercial metering to be provided for each User. It also sets out the requirement and procedures for metering.

VII. Data Registration Code. - This contains the details of all the data required by STU, which is to be provided by the Users and vice versa.

1.4

Scope of Grid Code. - 1. Grid Code defines the boundary between STU and Users and establishes the procedures for operation of facilities connected to the State Grid.

2.

All Users that connect with and/or utilize the State Grid are required to abide by the principles and procedures as laid down in the Grid Code in so far as they apply to that User.

3.

The Grid Code shall be enforced by STU. Users must provide STU reasonable rights of access, service and facilities necessary to discharge its responsibilities in the users premises and to comply with instructions as issued by STU to implement and enforce the Grid code.

1.5

Non-Compliance. - 1. If any User fails to comply with any provision of the Grid Code, the User shall inform the SLDC and Grid Code Review Panel without delay the reason for its non-compliance and shall remedy its non-compliance promptly.

2. SLDC shall bring the non-compliance to the notice of BERC.

3. Consistent failure to comply with the Grid Code provisions may lead to disconnection of the User's plant and /or facilities.

1.6

Code Responsibilities. - 1. In discharging its duties under the Grid Code, STU has to rely on information which Users shall supply regarding their requirements and intentions.

2. STU shall not be held responsible for any consequences that arise from its reasonable and prudent actions on the basis of such information.

1.7

Confidentiality. - 1. Under the terms of the Grid Code, STU will receive information from Users relating to their intentions in respect of their Generation or Supply businesses.

2. STU shall not, other than as required by the Grid Code, disclose such information to any person other than Central or State Government and BERC without the prior written consent of the provider of the information.

1.8

Dispute Settlement Procedures. - 1. In the event of any dispute regarding interpretation of any part/section of the Grid Code provision between any User and STU, the matter may be referred to the Commission for its decision. The Commission's decision shall be final and binding.

2. In the event of any conflict between any provision of the Grid Code and any contract or agreement between STU and Users, the provision(s) of the Grid Code will prevail.

1.9

Directive. - State Government may issue policy directives in certain matters as per Section 37 of the Electricity Act 2003. STU shall promptly inform the Commission and all Users of the requirement of such directives.

1.10

Compatibility with Indian Electricity Grid Code. - This Grid Code is consistent/compatible with the IEGC. However, in matters relating to Inter-State Transmission, if any provision of the State Electricity Grid Code are inconsistent with the provisions of the IEGC, then the provisions of IEGC as notified by CERC shall prevail.

1.11

The Board functioning as integrated Utility. - The functions of STU, SLDC, Generating Stations, Distribution Licensees shall be performed by the concerned officers authorised by the Board as long as it continues to function as an integrated Utility. Till such time the Board is functioning as an integrated utility, Ring fencing of SLDC shall be done separating manpower and accounts for SLDC functions.

1.12

Exemptions. - Any exemption from the provisions of the Grid Code shall be effective only after the approval of the BERC for which the agencies will have to file a petition in advance.

2. Glossary and Definition.

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In the Grid Code the following words and expressions shall, unless the subject matter or context otherwise requires or is inconsistent therewith, bear the following meanings:

ABT

Availability Based Tariff

Act

The Electricity Act, 2003 (Act No. 36 of 2003)

Active Energy

The electrical energy produced, flowing or supplied by an electric circuit during a time interval, being the integral with respect to time of the instantaneous power, measured in units of watt-hours or standard multiples thereof,

Active Power

The product of voltage and the in phase component of alternating current measured in units of watts and standard multiples thereof,

Apparatus

All the electrical apparatus like machines, fittings, accessories and appliances in which electrical conductors are used.

Apparent Power

The product of voltage and alternating current measured in unit of volt-amperes and standard multiples thereof,

Area of Supply

Area within which a Distribution Licensee is authorized by his licence to supply electricity.

Automatic Voltage

Regulatory (AVR)

A continuously acting automatic excitation system to control a Generating Unit terminal voltage.

Authority

Central Electricity Authority (CEA)

Auxiliaries

All the plant and machinery required for the generating unit's functional operation that do not form part of the generating unit.

Availability

"Fully Available" shall mean that the Generating Unit is available to its contracted capacity. In respect of the Transmission System, "Availability " shall mean the time in hours the Transmission System is capable of transmitting electricity at its rated voltage from the supply point to the delivery point and expressed as a percentage of Annual Availability.

Bulk Power Transmission

Agreements made between the STU and its H.T customers and between the Board and CTU.

Agreement Backing Down

Reduction of generation on instructions from SLDC / ERLDC by a generating unit under abnormal conditions.

Black start procedure

The procedure necessary to recover from partial or total blackout.

Black start capability

An ability in respect of a Black Start Station, for at least one of its generating units or CCGT units to start up from shut down and to energize a part of the system and be synchronized to the system upon instruction from the State Load Dispatch Centre, within two hours, without any external supply.

Black start stations

Generating stations having Black Start Capability.

Captive Power Plant

(CPP)

A Power Plant set up by any person to generate electricity primarily for his own use and includes a power plant set up by any co-operative society or association of persons for generating electricity primarily for use of members of such co-operative society or association.

CEA

Central Electricity Authority

Central Transmission

Utility (CTU)

Any Government Company which the Central Government may notify under sub Section (1) of Section 38 of the Electricity Act, 2003.

CERC Central

Electricity Regulatory Commission.

Connection

The electric power lines and electrical equipment used to effect a connection of a user's system to the Transmission System.

Connection conditions

Those conditions mentioned in Section 5 ("connection conditions") which have to be fulfilled before the User's

System is connected to the State Grid

Connection point

An electrical point of connection between the Transmission System and the User's System.

Consumer

Any person who is supplied with electricity for his own use by a licensee or the Government or by any other person engaged in the business of supplying electricity to public under the Electricity Act, 2003 or any other law for the time being in force and includes any person whose premises are for the time being connected for the purpose of receiving electricity with the works of a licensee, the Government or such other person, as case may be and shall include a person whose electricity supply has been disconnected.

Demand

The demand of Active Power in MW and Reactive Power in MVAR of electricity unless otherwise stated.

Demand control

Any of the following methods of achieving a load reduction:

- (a) Consumer Load Management initiated by Users.
- (b) Consumer Load reduction by Disconnection initiated by Users (other than following an instruction from Load Despatch Centre).
- (c) Consumer Load reduction instructed by the Load Despatch Centre
- (d) Automatic under Frequency Load Disconnection
- (e) Emergency manual Load Disconnection

Despatch

Operation control of an integrated electricity system involving operations such as:

- (a) Assignment of levels of output to specific Generating Plant or Load control devices to effect the most reliable and economical supply as the load vary.
- (b) The control of the operation of Extra High Voltage lines, associated sub stations and equipment.
- (c) The scheduling of various types of transactions with the electric utilities over the interconnecting Transmission Lines.

De-synchronize

The act of taking a generating unit off a system to which it has been synchronized.

Disconnection

The physical separation of Users or Consumers from the system.

Discrimination

The quality where a relay or protective system is enabled to pick out and cause to be disconnected only the faulty apparatus.

Distribution Licensees

A licensee authorized under Section 14 of the Act to operate and maintain a distribution system for supplying electricity to the consumers in his area of supply.

Distribution system

The system of wires and associated facilities between the delivery points on the transmission lines or the generating station connection and the point of connection to the installation of the consumers.

Drawal

The import / export of electrical energy from / to the grid

Earthing

Connecting the conducting parts of an equipment or machinery with the general mass of earth, in such a manner ensuring at all times an immediate discharge of energy without danger, by maintaining the same efficiently at earth's potential.

Earthing device

A means of providing connection between a conductor and earth being of adequate strength and capability.

EHV

Extra High Voltage equal to and greater than 66 kV

Frequency

The number of alternating current cycles per second (expressed in Hertz) at which the system is operating.

Generating company

Any company or body corporate or association or body of individuals, whether incorporated or not, or artificial juridical person, which owns or operates or maintains a generating station.

Generating station

Any station for generating electricity, including any building and plant with step-up transformer, switch-yard, switch gear, cables or other appurtenant equipment, if any used for that purpose and the site thereof, a site intended to be used for a generating station, and any building used for housing the operating staff of a generating station and where electricity is generated by water - power, includes, penstocks, head and tail works, main and regulatory reservoirs, dams and other hydraulic works, but does not in any case include any sub station.

Generating unit

The combination of an electric power generator and its prime mover and all of its associated equipment, which together constitutes a single generating machine.

Generating schedule

The despatch schedule of a generating station

BERC

Bihar Electricity Regulatory Commission

Grid

High Voltage backbone system of inter-connected Transmission Lines, Sub Stations and Generating Stations.

Grid Code

" Bihar Electricity Grid Code", a document describing the procedures and the responsibilities for planning and operation of Bihar Grid.

Grid Code Review Panel

or 'Panel'

The Panel with the functions set out in the Grid Code*

High Voltage or HV IEGC

Voltage greater than 440 V and lower than 33 kV. Indian Electricity Grid Code, a document describing the philosophy and the responsibilities for planning and operation of Indian Power System specified by the Central Electricity Regulatory Commission (CERC) in accordance with sub-section 1 (h) of Section 79 of the Act.

Indian Standards ("IS")

Those Standards and specifications approved by the Bureau of Indian Standards.

Inter-State Transmission System (ISTS)

Inter-State Transmission System includes :

- (a) Any system for the conveyance of electricity by means of a main Transmission Line from the territory of one State to another State;
- (b) the conveyance of electricity across the territory of an intervening State as well as conveyance within a State, which is incidental to such Inter-State transmission of electricity.
- (c) The transmission of electricity within the territory of a

State built, owned, operated maintained or controlled by the Central Transmission Utility.

Interconnecting

Transformer (ICT)

Transformer connecting EHV lines of different voltage systems.

Independent Power

Producer (IPP)

Power Station within the State owned by a generator who is not part of BSEB.

Inter tripping

(a) The tripping of circuit - breaker(s) by commands initiated from protection at a remote location independent of the state of local protection; or

(b) Operational intertripping.

Intra-State Transmission

System

Any system for transmission of electricity other than an Inter - State Transmission System.

Isolation

The disconnection of EHV / HV Apparatus from the remainder of the System in which that EHV / HV Apparatus is situated.

Lean Period

That period in a day for a few hours when the electrical power demand remains less than the average demand.

Licence

Any licence granted by BERC under provisions of the relevant laws in force

Load

The Active, Reactive or Apparent power as the context requires, generated, transmitted or distributed.

Load Factor

Load Factor is the ratio of the average power to the maximum demand. The load factor depends on the interval of time of the maximum demand and the period over which the average is taken.

{|

Load Factor = $\frac{\text{Units consumed in a given period}}{\text{Maximum demand} \times \text{No. of hours in the period}}$

| -

| Low Voltage or LV

| Voltage not exceeding 440 volts

| -

| Main protection

| Protection equipment or system expected to have priority in initiating either a fault clearance or an action to terminate an abnormal condition in a power system.

| -

| NTPC

| National Thermal Power Corporation Limited

| -

| Operating Margin

| Aggregate available capacity of generating station in the system on real time basis, which is over and above the operating level

to the maximum capacity of the generating units limited by technical parameters for short duration.

| -

| Operation

| A scheduled or planned action relating to the operation of a system.

| -

| Operational procedure

| Management instructions and procedures, both for the safety Rules and for the local and remote operation of plant and apparatus, issued in connection with the actual operation of plant and/or apparatus at or from a connecting site.

| -

| Open Access

| The non-discriminatory provision for the use of transmission lines or distribution system or associated facilities with such lines or system by any licensee or consumer or a person engaged in generation in accordance with the regulations specified by the Commission.

| -

| Outage

| A total or partial regulation in availability due to repair and maintenance of the Transmission or Distribution or Generation facility or defect in Auxiliary System.

| -

| Part Load

| The condition of a generating station which is loaded but is not running at its declared availability.

| -

| Partial shutdown

| A shutdown of a part of the system resulting in failure of power supply, either from external connections or from the healthy part of the system.

| -

| Peak period

| That period in a day when the electrical power demand is highest.

| -

| Person

| Any company or body corporate or association or body of individuals, whether incorporated or not, or artificial juridical person.

| -

| PGCIL

| Power Grid Corporation of India Limited

| -

| Planned outage

| An outage of generating plant or part of the Transmission system, or part of a User's System coordinated by SLDC.

| -

| Power factor

| The ratio of Active Power (kW) to Apparent Power (KVA).

|-

| Protection

| The scheme and Apparatus for detecting abnormal conditions on a system and initiating fault clearance or actuating signals or indications.

|-

| Rated MW

| The "rating plate" MW output of a Generating Unit, being that output up to which the generating unit is designed to operate.

|-

| Reactive Power

| The product of voltage and current and the sine of the phase angle between them measured in units of volt-amperes reactive and standard multiples thereof;

|-

| Regulating Margin

| The system voltage and frequency beyond which the system should not be operated.

|-

| Re-Synchronization

| The bringing of parts of the system which has gone out of synchronism with each other, back into synchronism.

|-

| Safety-Rules

| The Rules framed by the Users and the transmission licensee to ensure safety to persons working on plant /apparatus.

|-

| SLDC

| State Load Despatch Centre

|-

| Standing Instructions

| An instruction issued by SLDC to a generating company whereby, in specified circumstances, the generating company should take specified action, as though a valid dispatch instruction has been issued by SLDC.

|-

| Start-Up

| The action of bringing a generating unit from shutdown to synchronous speed.

|-

| State Transmission

Utility (STU)

| The utility notified by the Government under subsection (1) of Section 39 of the Electricity Act, 2003 and whose functions have been outlined under subsection (2) of Section 39 of the Electricity Act 2003.

|-

| Station Transformer

| A transformer supplying electrical power to the auxiliaries of a generating station, which is not directly connected to a

generating unit terminal.

|-

| Sub station

| Station for transforming or converting electricity for the transmission or distribution thereof and includes transformers, converters, switchgears, capacitors, synchronous condensers, structures, cable and other appurtenant equipment and any buildings used for that purpose and the site thereof.

|-

| Supervisory Control and
Data Acquisition or(SCADA)

| The communication links and data processing systems, which provide information to enable implementation of requisite supervisory and control actions.

|-

| Synchronized

| Those conditions where an incoming generating unit or system is connected to the bus bars of another system so that the frequencies and phase relationships of that generating unit or system as the case may be, and the system to which it is connected are identical.

|-

| System

| Any transmission and distribution system and / or transmission system, as the case may be.

|-

| Total system

| The transmission system and all user systems in Bihar.

|-

| Transmission licensee

| A licensee authorized to establish and operate transmission lines

|-

| Transmission lines

| All high pressure cables and overhead lines (not being an essential part of the distribution system of a licensee) transmitting electricity from a generating station to another generating station or a sub station, together with any step up and step-down transformers, switch-gear and other works necessary to and used for the control of such cables or overhead lines, and such buildings or part thereof as may be required to accommodate such transformers, switchgear and other works.

|-

| Transmission system

| The system consisting of high pressure cables and overhead lines of transmission licensee for transmission of electrical power from the generating station upto connection point / interface point with the distribution system. This shall not include any Part of ? the distribution system.

|-

| Under Frequency Relay

| An electrical measuring relay intended to operate when its

characteristic quantity reaches the relay settings by decrease in frequency.

| -

| User

| A term utilized in various Sections of Grid Code to refer to the persons using the Bihar Grid, as more particularly identified in each Section of the Grid Code. In the general conditions the term means any person to whom the Grid Code applies.

| }

Words and expressions used and not defined in this code but defined in the Acts shall have the meanings assigned to them in the said Acts. Expressions used herein but not specifically defined in this Code or in the said Acts but defined under any law passed by a competent legislature and applicable to the electricity industry in the State shall have the meaning assigned to them in such law. Subject to the above, expressions used herein but not specifically defined in this Code or in the Acts or any law passed by a competent legislature shall have the meaning as is generally assigned in the electricity industry.

3. Management of Grid Code.

- 3.1. Introduction.

1. The State Transmission Utility (STU) is required to implement and comply with the Bihar Electricity Grid Code and to carry out periodic review and amendments of the same with the approval of Bihar Electricity Regulatory Commission (BERC). A Review Panel shall be constituted by STU, as required in this Section, comprising of the representatives of the Users of the Transmission System.

2. No change in this Grid Code, however small or large, shall be made without being deliberated and agreed by the Grid Code Review Panel and thereafter approved by BERC. However, in an unusual situation where normal day to day operation is not possible without revision of some clauses of Grid Code, a provisional revision may be implemented before approval of BERC is received, but only after discussion at a special Review Panel Meeting convened on emergency basis. BERC should promptly be intimated about the provisional revision. BERC may issue directions requiring STU to revise the Grid Code accordingly as may be specified in those directions and STU shall promptly comply with any such direction.

3.2

Objective. - The objective of this Section is to define the method of management of Grid Code documents, implementing any changes / modifications required and the responsibilities of the constituents (Users) to effect the change.

3.3

Grid Code Review Panel. - 1 .The Chairperson of the Grid Code Review Panel shall be an Engineer of the STU not below the rank of Chief Engineer. The Member Secretary of the Panel shall also be nominated by STU. The Grid Code Review Panel shall consist of the following members in addition to above on the recommendations of the heads of the respective organizations:

(a)

One member from SLDC (head of SLDC)

(b)

One Chief Engineer (Generation) of Bihar State Electricity Board.

(c)

One member from Transmission licensee in the State other than STU

(d)

One representative at senior executive level from system operation from National Thermal Power Corporation Limited (NTPC),

(e)

One member at senior level from ERPC Secretariat

(f)

One representative at senior executive level from Eastern Regional Load Despatch Centre (ERLDC)

(g)

One representative at senior executive level from Distribution Licensee.

(h)

One representative at senior executive level from each of the generating companies feeding not less than 50 MW to the Grid in the State.

(i)

One representative from all Captive Power Plants (CPPs), which are in parallel operation with Bihar State Grid, on rotation basis.

(j)

One representative from all the generating companies of small generating stations of less than 50 MW capacity on rotation basis.

2. Any other member can be co-opted as a member of the panel when directed by BERC.

3. The functioning of the Committee shall be co-ordinated by STU. The Member Secretary nominated by STU shall be the convener.

4.

STU shall inform all the Users, the names and addresses of the Review Panel, Chairperson and the Member Secretary at least 15 days before the first Review Panel meeting. Any subsequent changes shall also be informed to all the Users by STU. Similarly, each user shall inform the names and designations of their representatives to the Member Secretary of the Review Panel, at least three days before the first Review Panel meeting, and shall also inform the Member Secretary in writing regarding any subsequent changes.

3.4

Functions of the Review Panel. - The functions of the Review Panel are as follows:

(a)

Review and Maintenance of the Grid Code and its working.

(b)

Consideration of all requests for review made by any user and make recommendations for changes in the Grid Code together with reasons for such changes.

(c)

Provide guidance on interpretation and implementation of the Grid Code.

(d)

Examination of the problems raised by any user as well as resolution of the problems.

(e)

Ensuring that the changes / modifications proposed in the Grid Code are consistent and compatible with Indian Electricity Grid Code (IEGC). The Review Panel may hold any number of meetings as required subject to the condition that at least one meeting shall be held in every three months. Sub-meetings may be held by STU with the Users to discuss individual requirements and with groups of Users to prepare proposals for Co-ordination Committee's consideration.

3.5

Review and Revisions. - 1. The Users seeking any amendment to the Grid Code shall send written requests to the Member Secretary of the Review Panel. If the request is sent to BERC directly, the same shall be forwarded to STU. STU shall examine the proposed changes / modifications in line with IEGC stipulations and circulate the same along with its comments to all Review Panel members for their written comments within a reasonable time frame. Whenever it is observed that a certain clause of Grid Code is not consistent with the IEGC, then the same will be discussed in the Review Panel and the clause will be revised to make it consistent with IEGC.

2. All the comments received shall be scrutinized and compiled by STU. These along with STU's comments shall be sent to all the members for their response for the proposed change / modification. If necessary, STU shall convene a meeting of the Review Panel for deliberations. The Member Secretary shall present all the proposed revisions of the Grid Code to the Review Panel for its consideration.

3. Based on the response received, STU shall finalise its recommendation regarding the proposed modification / amendment and submit the same along with all the related correspondence to BERC for approval.

4. STU shall send the following reports to the BERC at the conclusion of each review meeting of the Committee:

(a)

Report on the outcome of such review.

(b)

Any proposed revision to the Grid Code.

(c)

All written representations and objections submitted by the Users at the time of review.

5.

All revisions to the Grid Code require the approval of BERC. STU shall publish revisions to the Grid Code, after the approval of BERC. STU may submit proposals for relaxation in such cases where Users have difficulties in meeting the requirements of the Grid Code.

6.

Any change from the previous version shall be clearly marked in the margin. In addition, a revision sheet shall be placed at the front of the revised version noting the number of every changed sub-section, together with reasons for such change.

7.

STU shall maintain copies of the Grid Code with the latest amendments and shall make it available at a reasonable cost to any person requiring it. This may also be made available on the website. The STU shall keep an up to date list of recipients of all the copies of the Grid Code, if found necessary to ensure that the latest version of Grid Code is reached to all the relevant recipients.

8.

The Commission, may, on the application of the Users or otherwise, call the emergency meeting of the Review Panel as and when the situation so dictates and make such alterations or amendments in the Grid Code as it thinks fit.

Chapter 2

Planning Code

4. System Planning.

- 4.1. Introduction -

1. This Section specifies the methods for data submission by Users to STU for planning and development of the State Transmission System. This Section also specifies the procedures to be adopted by STU in the planning and development of the State Transmission System.

2. Requirement for reinforcement or extension of the State Transmission system arise due to many reasons of which a few are mentioned below:

(i)

Development on a User's system already connected to the State Transmission System.

(ii)

Introduction of a new connection point between the User's system and the State Transmission System.

(iii)

Evacuation system for generating stations within or outside State

(iv)

Reactive compensation.

(v)

A general increase in system capacity due to addition of generation or system load.

(vi)

Transient or steady state stability considerations

(vii)

Cumulative effect of any of the above

3. The reinforcement or extension of the State Transmission System may involve work at an entry or exit point (connection point) of a User to the State Transmission System. Since development of all User's systems must be planned well in advance to ensure consents and way leaves to be obtained and detailed engineering design / construction work to be completed, STU will require information from Users and vice versa. To this effect, the planning code imposes time scale for exchange of necessary information between STU and Users.

4.2

Objective. - This Section formulates the procedures for the 'System Planning' to enable STU in consultation with the Users, to evolve an efficient, coordinated, secure and economical State Transmission System to satisfy requirements of

future demand.

4.3

Planning Policy. - 1. STU would develop a prospective transmission plan for next 10 years for the State Transmission System. These plans shall be updated every year to take care of the revisions in load projections and generation capacity additions. The perspective plans shall be submitted to the Commission for approval.

2. STU shall carry out network studies and review fault levels for planning system strengthening and augmentation.

3. STU shall follow the following steps in planning:

(i)

Forecast the demand for power within the area of supply based on the forecasts provided by Distribution Licensees. These shall include details of demand forecasts, data methodology and assumptions on which forecasts are based. These forecasts would be annually reviewed and updated, and also whenever major changes are made in the existing system.

(ii)

Prepare a proposal for the requirement of generation for the State to meet the load demand as per the forecast taking into account the existing contracted generation resources.

(iii)

Prepare a transmission plan for the State Transmission System compatible with the above load forecast and generation plan. VAR compensation needed will also be included.

(iv)

Necessary load flow studies shall be carried out for short circuit and transient stability study and relay coordination study for transmission system planning.

(v)

STU shall be responsible to prepare and submit a long term (10 years) plan to BERC for generation expansion and transmission system expansion to fully meet both energy and peak demand.

(vi)

STU shall extend support to Central Transmission Utility (CTU) to finalise the annual planning corresponding to a 5 years forward term for identification of major Inter-State Transmission System including inter-regional schemes which shall fit in with the long term plan developed by CEA.

4. All Users shall supply to STU, the planning data prescribed in Appendix A and Appendix B of Data Registration Code within 3 months from the effective date of the Grid Code and thereafter such data shall be furnished by 31st March every year to enable STU to formulate/ finalise the updated plan by 30th September each year for the next 5 years.

4.4

Planning Criterion. - 1. The planning criterion shall be based on the security philosophy on which both Inter State Transmission System (ISTS) and the State Transmission System (STS) have been planned. The security philosophy shall be as per the Transmission Planning criteria and guidelines as given by Central Electricity Authority (CEA).

2. The State Transmission System planning and generation expansion planning shall be in accordance with the provisions of the planning criterion as per IEGC Clause 3.5.

4.5

Planning responsibility. - 1. The primary responsibility of load forecasting within distribution licensee's area of supply rests with the respective Distribution Licensee. The Distribution Licensee shall determine peak load and energy forecast of their areas for each category of loads for each of the succeeding 5 years and submit the same annually by 31st March to STU along with details of demand forecasts, data, methodology and assumptions on which forecasts are based along with their proposals for transmission system augmentation. The load forecasts shall be made for each of the prevalent as well as proposed inter connection points between STU and Distribution Licensees and shall include annual peak load and energy projections.

2. Generating stations shall provide their generation capacity to STU for evacuating power from their power stations for each of the succeeding 5 years along with their proposals for augmentation of transmission proposals and submit the same annually by 31st March to STU.

3. The planning for strengthening the State Transmission System for evacuation of power from outside State Stations shall be initiated by STU.

4.6

Planning data. - 1. To enable STU to conduct system studies and prepare perspective plans for electricity demand, generation and transmission, the Users shall furnish data to STU from time to time as detailed under Data Registration Section as under:

(a)
Standard Planning Data (Generation)/Standard Planning Data (Distribution) as per Appendix A.

(b)
Detailed Planning Data (Generation)/Detailed Planning Data (Distribution) as per Appendix B.

2. To enable the Users to co-ordinate planning design and operation of their plants and systems with the State Transmission System, they may seek certain data of transmission system as applicable to them, which the STU shall supply from time to time as detailed under Data Registration Section categorized as;

(a)
Standard Planning Data (Transmission) as per Appendix A.

(b)
Detailed Planning Data (Transmission) as per Appendix B.

4.7

Implementation. - The actual programme of implementation of transmission lines, inter - connecting transformers, reactors/capacitors and other transmission elements will be determined by STU in consultation with the concerned Users. The completion of these works in the required time frame shall be ensured by STU through the concerned users. STU/SLDC may periodically inform RLDC of their plans as well as stages of implementation.

5. Connection Conditions.

- 5.1. Introduction -

This Section specifies the technical, design and operational criteria which shall be complied with by every User connected or seeking connection to the State Transmission System.

Any user seeking connection to or connected with the Intra-State System shall comply with the CEA Regulations 2007 "Technical Standards for connectivity to the Grid".

This applies to all users seeking connection or connected to the Intra-State System.

5.2

Objective. - The objective of this Section is to ensure the following:

(i)
All Users or prospective Users are treated in a non-discriminatory manner

(ii)
Any new or modified connection when established shall neither suffer unacceptable effects due to its connection to State Transmission System nor impose any unacceptable effect on the system of existing Users and new connection shall not suffer adversely due to existing Users.

(iii)
A system of acceptable quality is ensured by specifying the required minimum standards for the design and operational criteria to assist the Users in their requirement to comply with the licence obligations.

(iv)
The ownership and responsibility for all equipments is clearly specified in the "Site Responsibility Schedule" for every site, where a connection is made.

5.3

Procedure for application for connection to Transmission System. - The procedure for any new connection or modification of an existing connection with the State Transmission System shall consist of following:

1. The User shall meet all necessary conditions as specified in the State Grid Code and submit the application to STU containing all the information as may be specified.

2. STU shall make a formal offer within 60 days of the receipt of the application. The offer shall specify and take into account any work required for the extension or reinforcement of the State Transmission System necessitated by the applicant's proposal and for obtaining any consent necessary for the purpose. STU shall process the application in accordance with the CEA Regulations on Grid Connectivity

3. If the specified time limit for making the offer against any application is not adequate, STU shall make a preliminary offer within the specified time indicating the extent of further time required for detailed analysis.

4. Any offer made by STU shall remain valid for a period of 60 days and unless accepted before the expiry of such period, shall lapse thereafter.

5. In the event of offer becoming invalid or not accepted by the applicant, STU shall not be bound to consider any further application from the same applicant within 12 months unless the new application is substantially different from the original application.

6. The applicant shall furnish the Detailed Planning Data as per Appendix-B.

5.4

Rejection of application. - STU shall be entitled to reject any application for connection to or use of the State Transmission System due to the following reasons apart from others as considered reasonable:

(i)

If such proposed connection is likely to cause breach of any provision of its Licence or any provision of the Grid Code or any provision of IEGC or any criteria or covenants or deeds or Regulations by which STU is bound.

(ii)

If the applicant does not undertake to be bound, in so far as applicable, by the terms of the Grid Code.

(iii)

If the applicant fails to give confirmation and undertakings according to this Section.

(iv)

The reason for rejection of connection to or use of the State Transmission System will be communicated to the applicant by STU within 60 days of the receipt of application.

5.5

Connection Agreement. - 1. A connection agreement, or the offer for a connection agreement shall include (but not limited) within its terms and conditions the following:

(i)

A condition requiring both agencies to comply with the Grid Code.

(ii)

Details of connection technical requirements and commercial arrangements

(iii)

Details of any capital related payments arising from necessary reinforcement or extension of the system, data communication, RTU etc and demarcation of the same between the concerned parties.

(iv)

A Site Responsibility Schedule (Appendix-D).

(v)

General Philosophy, Guidelines etc on protection and telemetry

2. Model Connection agreement /-This shall be prepared by STU and submitted to the Commission for approval.

5.6

Site responsibility schedule. - 1. For every connection to the State Transmission System for which a connection agreement is required, STU shall prepare a schedule of equipment with information supplied by the respective Users. This schedule, called 'Site Responsibility Schedule' shall indicate the following for each item of equipment installed at the connection site.

(i)

Ownership of the equipment

(ii)

Responsibility for control of equipment

(iii)

Responsibility for maintenance of equipment

(iv)

Responsibility for operation of equipment

(v)

Responsibility for all matters relating to safety of persons and site.

(vi)

Management of the site.

2. The format to be used in the preparation of Site Responsibility Schedule is given in Appendix - D.

5.7

System Performance. - 1. All equipment connected to the State Transmission System shall be of such design and construction that enable STU to meet the requirement of standards of performance specified by BERC under Section 57 of the Electricity Act, 2003.

2. Installation of all electrical equipment shall comply with IE Rules, 1956 which are in force for time being and will be replaced by new Rules made under Electricity Act, 2003.

3. For every new/modified connection sought the STU shall specify the connection point, technical requirements and the voltage to be used, along with metering, telemetering and protection requirements as specified in the Metering Code and Protection Code.

4. Insulation coordination of the User's equipment shall conform to the applicable values as specified by STU from time to time out of those applicable as per Indian Standards / Code of Practices. Rupturing capacity of the switch gear shall not be less than that specified by STU from time to time.

5. Protection schemes and metering schemes shall be as detailed in the Protection Code and Metering Code.

6. The State Transmission System rated frequency shall be 50.00 Hz and shall be regulated by the provisions of IEGC as given below:

Target Range

Upper limit

-

50.20 Hz

Lower limit

-

49.50 Hz

7. The Users shall be subject to the Grid discipline prescribed by SLDC and ERLDC.

8. The variation of voltage at the interconnected point should not be more than voltage range specified below:

Nominal (KV)

% limit of variation

Maximum (KV)

Minimum (KV)

400

+5%/- 5%

420

380

220

+/-10%

245

198

132

+/-10%

145

122

5.8

Connection Points/interface points. - 1. State Generating Stations (SGS)

(i)

Voltage may be 220/132/33 KV or as agreed with STU.

(ii)

Unless specifically agreed with STU, the Connection point with generating station shall be the terminal isolator provided just before the outgoing gantry of the feeders.

(iii)

SGS shall operate and maintain all terminals, communication and protection equipments provided within the generating station.

(iv)

The provisions for the metering between generating station and STU system shall be as per the Metering Code.

(v)

Respective Users shall maintain their equipments from the out going feeders' gantry onwards emanating from generating station'

(vi)

All entities embedded within BSEB system and interfacing the Intra/ State Transmission System shall provide adequate and reliable communication facility so that SLDC is able to record in its SCADA system the MW/ MVAR flows, bus voltages at all the interface points with the intra-state system.

2. Distribution Licensee. - (i) Voltage may be LV side of power transformer i.e. 33 KV or 11 KV or as agreed with STU. For EHV consumers directly connected to transmission system, voltage may be 220 KV/132 KV

(ii)

Unless specifically agreed with Distribution Licensee, the Connection point with STU shall be the terminal isolator provided just before the outgoing gantry of the feeder to Distribution Licensee or individual EHV consumer as the case may be, from STU sub-station

(iii)

STU shall operate and maintain all terminals, communication and protection equipments provided within its sub-station. The provisions for the metering between STU and Distribution Licensee systems shall be as per the Metering Code. Respective Users shall maintain their equipment beyond the out going gantry of feeders emanating from STU sub-station onwards.

3. Eastern Regional Transmission System. - The Connection, protection scheme, metering scheme and the voltage shall be in accordance with the provisions of IEGC.

4. Independent Power Producers (IPPs), Captive Power Plants (CPPs), Extra High Voltage (EHV) Consumers and Open Access Customers. -

(i)

Voltage may be 220/132/33 KV or as agreed with STU.

(ii)

When IPPs, CPPs, EHV Consumers or the Open Access Customers own sub-stations, the Connection point shall be the terminal isolator provided just before the gantry of outgoing/incoming feeder in their premises.

Chapter 3

System Operation Code

6. Operation Planning.

- 6.1. Introduction -

This Section describes the process by which the SLDC carries out the operational planning and demand control procedures to permit reduction in demand for any reason.

6.2

Objective. - Operational Planning is aimed at integrated, economic and reliable operation of the State Grid.

Operating Policy:

(i)

Overall real time operation of the State Grid shall be supervised by the SLDC and the STU in accordance with Grid Code.

(ii)

Distribution Licensee shall monitor grid operation of its distribution system and coordinate with the SLDC.

(iii)

All State entities shall comply with the operation guidelines specified hereinafter and co-ordinate with each other, for deriving maximum benefits from the integrated operation and for equitable sharing of obligations.

(iv)

A set of detailed internal operating procedures for the State Grid shall be developed and maintained by the SLDC in consideration with the entities and same shall be consistent with Grid Code. The control rooms of the SLDC, Area LDC, Generating Stations, and EHV substations and any other control centers of all the entities shall be manned round the clock by qualified personnel with adequate training.

6.3

System security. - 1. All Users shall operate their respective power system and generating stations in synchronization with each other at all times so that the whole State Transmission System operates as a synchronized system as well as integrated part of Eastern Region Grid. STU shall operate the Inter-State links so that Inter State transfer of power can be achieved smoothly when required.

2. No part of the State Transmission System shall be isolated from the integrated grid except under the following conditions;

(i)

Emergency situations that may result in total grid collapse.

(ii)

Isolation of the system to prevent serious damage to important/costly equipment.

(iii)

When such isolation is specifically instructed by SLDC.

(iv)

On operation of under frequency / islanding scheme as approved at Eastern Region level.

3. Complete synchronization shall be restored as soon as conditions permit. The restoration process shall be supervised by SLDC.

4. The 132 KV and above transmission lines (except radial lines which do not affect the operation of the Grid) and the inter connecting power transformers should not be opened or removed from service without instruction or prior clearance from the SLDC. Under emergencies where prior clearance from SLDC is not possible, it should be intimated to SLDC at the earliest possible time after the incident and get the clearance, while bringing back these lines into service.

5. Any tripping, whether manual or automatic of transmission lines by 132 KV and above or power transformers of 132 KV class and above and 50 MVA and above shall be promptly reported to the SLDC at the earliest along with the reasons for such tripping and the likely time required for restoration. The information / data including that down loaded from disturbance recorder, sequential event recorder etc. required for the purpose of analysis shall be sent to SLDC. For restoration of tripped equipment / line, SLDC shall be informed and get its clearance.

6. All generating units of 200 MW and above in case of thermal and 10 MW and above in case of hydro which are synchronized with the grid irrespective of their ownership shall have their governors in normal operation at all times. If any generating unit of over 50 MW size is required to be operated without its governor in normal operation, the ERLDC through SLDC shall be immediately intimated about the reason and duration of such operation. The exemption from free governor mode operation in respect of run of river hydro stations without any pondage, steam turbine of thermal and gas based power stations not having free governor mode facility shall be obtained from CERC under clause 1.6 of IEGC.

7. Facilities available with / in Load limiter, Automatic Turbine Run-up System (ATRS), Turbine Supervisory Coordinated Control System etc., shall not be used to by pass the normal governor action in any manner. No dead bands and time delays shall be deliberately introduced.

8. All generating units operating in Free Governor Mode Operation (FGMO) at or upto 100% of their maximum continuous rating shall normally be capable of (and shall not be prevented from) picking upto 5% extra load, more than the declared maximum continuous rating, for atleast five minutes or within the technical limits specified by the manufacturers, when the frequency falls due to a system contingency. In case any generating unit of 50 MW and above size does not meet this requirement for any period, the generating company should intimate the same to SLDC along with reasons thereof. Any generating unit not capable of complying with above provisions either due to not having requisite facilities or otherwise shall seek exemption from Central Electricity Regulatory Commission (CERC) under clause 1.6 of IEGC.

9.

In case frequency falls below 49.5 Hz, all partly loaded Generating Units shall pick up additional load at a faster rate, according to their capability. SLDC in consultation with ERLDC and Distribution Licensees shall prepare a plan for automatic load relief during the low frequency conditions. In case frequency rises to 50.5 Hz or higher, neither any generating unit shall be synchronized with the Grid nor shall generation at any generating station (irrespective of type or ownership) be increased without obtaining approval from SLDC.

10.

Except under an emergency, or to prevent an imminent damage to costly equipment, no User shall suddenly

decrease/increase its generation without prior intimation to the SLDC. Similarly, no User shall cause a sudden decrease/increase in its load due to imposition/lifting of power cuts etc., without prior intimation to and consent of SLDC, particularly when frequency is deteriorating.

11.

All Generating Units shall normally have their Automatic Voltage Regulators (AVRs) in operation, with appropriate settings. In particular, if a Generating Unit of over 50 MW capacity is required to be operated without its AVR in service, the same should be operated only after prior concurrence of SLDC.

12.

Each Generating Unit must be fitted with a turbine speed governor having an overall droop characteristic within the range of 3% to 6%, which shall always be in service.

13.

State Generating Stations and other generating stations connected to the Grid shall follow the instructions of SLDC for backing down/boxing up (ramping-down) and shutting down the generating unit(s). SLDC shall provide the certificate for the period of the backing down/boxing up or shutting down for the purpose of computing the deemed generation, if required.

14.

Various steps shall be taken for frequency management and voltage management so as to ensure system security from these considerations.

6.4

Demand Estimation. - 1. The Distribution Licensee shall formulate a short-term demand forecast considering the previous financial year as base and projecting the demand for the succeeding 5 years.

2. It shall be the responsibility of all Distribution Licensees to fully cooperate with STU in preparation of demand forecast for the entire State.

3.

The long term demand estimation / load forecast (for more than 1 year) shall be done by STU and SLDC shall be provided with a copy of the same as and when it is finalized.

SLDC shall inform its load projections (both peak and off - peak) in ERPC forum for three months, to be reviewed every month ahead.

4. The Distribution Licensees shall provide to the STU and SLDC their estimates of demand for each inter connection point for the next financial year by 31st January of each year. Distribution Licensees shall also provide daily demand for the month ahead at each inter connection point by 25th of the month.

5. Based on the data furnished by the Distribution Licensees, STU shall make monthly peak and lean period demand estimates for year ahead and daily peak and lean period demand estimates for the month ahead and furnish the same to SLDC.

6.

The Distribution Licensee shall provide to SLDC on day ahead basis, at 9.00 hrs, each day their estimated demand for each 15 minute block for the ensuing day. The Distribution Licensee shall also provide to SLDC estimates of loads that may be shed, when required, in discreet blocks with details of arrangement of such load shedding.

7. The SLDC would update demand forecast (in MW as well as kWh) on quarterly, monthly, weekly and ultimately on daily basis which would be used in the day - ahead scheduling.

6.5

Demand Control. - 1. Automatic load shedding shall be resorted to by means of installation of the Under Frequency Relays at the sub stations of the STU as per the directions of the SLDC to preserve the overall integrity of the power system. The number and size of the discrete blocks using Automatic Under Frequency Relays for Load Shedding shall be determined on rotational basis in consultation with every Distribution Licensee. The frequency settings of these relays shall be coordinated in consultation with the RLDC.

2. Whenever restoration of large portions of the total demand disconnection effected by the automatic load shedding is not possible within a reasonable time, the SLDC shall implement additional disconnection manually, to restore an equivalent amount of demand disconnected automatically.

Each Distribution Licensee shall help the SLDC in identifying such load blocks.

Load shed by the operation of automatic load shedding devices shall not be restored without specific directions from the

SLDC.

3. Planned manual disconnections shall be implemented by the SLDC when there is a shortfall in generation, or constraints in Transmission System, or reduction of imports through external connection etc., requiring demand control to control the over-drawl of power from Inter State Generating State (ISGS) when the system frequency falls below 49.5 Hz. In such cases a rotational load shedding scheme shall be adopted to ensure equitable treatment for all consumers as far as practicable.

4. Emergency Manual Disconnection to deal with unacceptable voltage and frequency levels etc. shall be implemented by the SLDC when loss of generation, mismatch of generation with the demand, constraints in the transmission system, over-drawl from the grid in excess of respective schedule affecting the frequency of the regional grid below 49 Hz, requiring load shedding at short notice or no notice, to maintain a regulating margin.

5. These control measures shall not be withdrawn till the system frequency improves and when the SLDC issues such instructions after review of the situation.

6.6

Load Crash. - 1. In the event of load crash in the system due to weather disturbance or any other reasons, the situation would be controlled by SLDC by the following methods in descending priorities:

- i. Lifting of the load restrictions, if any
- ii. Exporting the power to neighbouring Regions/States provided the same does not endanger the security of the ISTS
- iii. Backing down of thermal stations with a time lag of 5-10 minutes for short period in merit order.
- iv. Closing down of hydel units (subject to non spilling of water and effect on irrigation) keeping in view the inflow of water into canals and safety of canals/hydel channels.

2. While implementing the above, the system security aspects should not be violated as per provisions in Section 5.2 of IEGC and Section 6.3 of the Grid Code.

7. Outage Planning.

- 7.1. Introduction -

This Section describes the procedure for preparation of outage schedules for the elements of the State Grid in a coordinated and optimal manner keeping in view the State Grid operating conditions and the balance of generation and load.

7.2

Objective. - The objective of this Section is to define the process which will allow STU to optimise transmission outages with SGS (other than CPP) and Distribution Licensees, Outages in co-ordination with outage planning of regional system while maintaining system security to the extent possible.

7.3

Outage Planning Process. - 1. Each User shall provide their outage programme for ensuing financial year to SLDC for preparing an overall outage plan for the State Transmission System as a whole. SLDC shall be responsible for analyzing the outage schedules of all Users including SGS, Distribution Licensees, STU and Transmission Licensee(s) schedules for outage of Transmission network and preparing a draft annual Outage Plan for the State Transmission System in coordination with the Outage Plan prepared for the region by ERLDC. The Users shall furnish the information to SLDC as per Appendix-C.

2. SLDC is however authorised to defer the outage in case of any of the following events:

- ? Major grid disturbance
- ? System Isolation
- ? Black out in the State
- ? Any other event in the system that may have an adverse impact on system security by the proposed outage

3. Each User shall obtain approval of SLDC, prior to availing the Outage. SLDC while permitting any circuit for outage shall issue specific code. Similarly, no inter user boundary circuits shall be connected back to the State Transmission System without specific code/approval by SLDC.

This restriction shall however not be applicable to individual Generating Unit(s) of a CPP.

7.4

Annual Outage Planning. - 1. Scheduled outage of power stations of capacity 25 MW and above as notified by SLDC from time to time, will be subject to annual planning.

2. Scheduled outage of power station of 50 MW and above and EHV lines as notified by ERLDC, will also be subject to

annual planning of ERPC Secretariat in co-ordination with SLDC.

3. SGS (except CPPs) connected to the State Grid shall furnish their proposed Outage programme for the next financial year in writing by 15th November each year.

4. SGS Outage programme shall contain details like identification of unit, reason for outage, generation availability affected due to such outage, outage start date and duration of outage. SLDC shall review the outage programme received from SGS on monthly basis to chalk out the outage of the State Transmission System.

5. SLDC shall also obtain from STU, the proposed outage programme for Transmission lines, equipments and sub-stations etc. for next financial year by 15th November each year. STU outage programme shall contain identification of lines/ substations, reasons for outage, outage start date and duration of outage.

6. Scheduled outage of power stations and EHV transmission lines affecting regional power system shall be affected only with the approval of ERPC in coordination with SLDC.

7. Scheduled outage of power stations of capacity 25 MW and above, of all EHV lines and HV lines forming interconnection between two EHV substations (and those notified as such by SLDC) shall be approved by SLDC, 24 hours in advance based on prevalent operating conditions.

8. In respect of scheduled outage referred in this Section a calendar shall be formulated in respect of annual outage planning for the ensuing financial year. Such outage plan shall be deliberated and finalised in the meeting of the Operation and Co-ordination Committee (OCC). to be constituted by SLDC at the State level.

a. SLDC shall submit the proposed annual outage plan of the next financial year to ERPC Secretariat by 30th November of preceding year.

b. Final annual outage plan shall be communicated to ERPC Secretariat by 15th January of each year.

7.5

Availing of shutdowns schedule. - SLDC would review on daily basis the outage schedule for the next two days and in case of any contingency or conditions described in Section 5.7.4(g) of the IEGC, defer any planned outage as deemed fit clearly stating the reasons thereof. The revised dates in such cases would be finalized in consultation with the User.

8. Contingency Planning.

- 8.1. Introduction -

This Section describes the steps in the recovery process to be followed by all Users in the event of total or partial blackouts of the State Transmission System or Regional System.

8.2

Objective. - The objective of this Section is to define the responsibilities of all Users to achieve the fastest recovery in the event of the State Transmission System or Regional System blackout, taking into account essential loads, generator capabilities and system constraints.

8.3

Contingency Planning Procedure. - 1. SLDC shall be prepared to efficiently handle the following types of contingencies and restoration of system back to normal:

? Partial system black out in the State due to multiple tripping of the Transmission lines emanating from power stations/sub-stations

? Total black out in the State/Region

? Synchronisation of system islands and system split

2. In case of partial black out in the system/State, priority is to be given for early restoration of power station units, which have tripped. Start up power for the power station shall be extended through shortest possible route and within shortest possible time from adjoining substation/ power station where the supply is available. Synchronising facility at all power stations and 220 kV sub-station having interconnection with ISTS shall be available.

3. In case of total regional black out, SLDC In-charge shall co-ordinate and follow the instructions of ERLDC for early restoration of the entire grid. Startup power to the thermal stations shall be given by the hydel stations or through interstate supply, if available. All possible efforts shall be made to extend the hydel supply to the thermal power stations through shortest transmission network so as to avoid high voltage problem due to low load conditions. For safe and fast restoration of supply, SLDC shall formulate the proper sequence of operation for major generating units, lines, transformers and load within the State in consultation with ERLDC. The sequence of operation shall include opening, closing/tripping of circuit breakers, isolators, on-load tap-changers etc.

8.4

Restoration Procedure. - 1. Detailed procedure for restoration of the State Transmission System shall be prepared by SLDC for the following contingencies and shall be in conformity with the System Restoration Procedure of the Eastern Region prescribed under IEGC.

- Total System Black out
- Partial System Blackout
- Synchronisation of System Islands and System Split

2. The restoration process shall take into account the generator capabilities and the operational constraints of Regional and the State Transmission System with the object of achieving normalcy in the shortest possible time. All Users should be aware of the steps to be taken during major Grid Disturbance and system restoration process.

8.5

Special Considerations. - 1. During restoration process following the State Transmission System or Regional System blackout conditions, normal standards of voltage and frequency need not be applied and left to the discretion of the SLDC.

2. Distribution Licensees with essential loads shall separately identify non-essential components of such loads, which may be kept off during system contingencies. They shall also draw up an appropriate schedule with corresponding load blocks in each case. The nonessential loads can be put on only when system normalcy is restored, as advised by SLDC.

3. All Users shall pay special attention to carry out the procedures so that secondary collapse due to undue haste or inappropriate loading is avoided.

4. Despite the urgency of the situation, careful, prompt and complete logging of all operations and operational messages shall be ensured by all Users to facilitate subsequent investigation into the incident and the efficiency of the restoration process. Such investigation shall be conducted promptly after the incident and placed before the Grid Code Review Panel in its next meeting.

9. Cross Boundary Safety.

- 9.1. Introduction -

This Section specifies the requirements for safe working practices for maintenance of equipment associated with cross boundary operations. It lays down the procedure to be followed when work is required to be carried out on electrical equipment that is connected to another User's system.

9.2

Objective. - The objective of this Section is to achieve an agreement and consistency on the principles of safety as prescribed in the Indian Electricity Rules 1956 which are in force for time being and will be replaced by the Rules made under Electricity Act, 2003 when working across the inter user boundary between one User and another User.

9.3

Designated Officers. - STU and all Users shall nominate suitable authorized persons to be responsible for the co-ordination of safety across their boundary. These persons shall be referred to as Designated Officer(s) (or control person(s)).

9.4

Procedure. - 1. STU shall issue a list of Designated Officers (names, designations and telephone numbers) to all Users who have a direct inter user boundary with STU or other Users. This list shall be updated promptly whenever there is change of name, designation or telephone number.

2. All Users with a direct inter user boundary with STU or other User system shall issue a similar list of their Designated Officers to STU or other User(s), which shall be updated promptly whenever there is a change in the list.

3. Whenever work across a cross boundary / an inter-user boundary is to be carried, the Designated Officer of the User including STU itself, wishing to carry out work shall personally contact the other relevant Designated Officer. If the Permit to Work (PTW) cannot be obtained personally, the Designated Officers shall contact through telephone and exchange Code words to ensure correct identification of both agencies.

4. Should the work extend over more than one shift, the Designated Officer shall ensure that the relief Designated Officer is fully briefed on the nature of the work and the code words in operation.

5. The Designated Officer(s) shall co-operate to establish and maintain the precautions necessary for the required work to be carried out in a safe manner. Both the established isolation and the established earth shall be locked in position, where such facilities exist, and shall be clearly identified.

6. Work shall not commence until the Designated Officer of the User including STU itself, wishing to carry out the work, is satisfied that all the safety precautions have been established. This Designated Officer shall issue agreed safety documentation (PTW) to the working party to allow work to commence. The PTW in respect of specified EHV lines and other interconnections shall be issued with the consent of SLDC.

7. When work is completed and safety precautions are no longer required, the Designated Officer who has been responsible for the work being carried out shall make direct contact with the other Designated Officer to return the PTW and for removal of those safety precautions. Return of PTW in respect of specified EHV lines and interconnections shall be informed to SLDC.

8. The equipment shall only be considered as suitable for connecting back to service when all safety precautions are confirmed as removed, by direct communication using code word contact between the two Designated Officers, and after ensuring that the return of agreed safety documentation (PTW) from the working party has taken place.

9. STU shall develop an agreed written procedure for inter-user boundary safety and continually update it.

10. Any dispute concerning inter user Boundary Safety shall be resolved at the level of STU, if STU is not a party. In case STU is a party, the dispute shall be referred to BERC for resolution of the dispute.

9.5

Special Consideration. - 1. For inter-user boundary between STU and other User's circuits, all Users shall comply with the agreed safety Rules, which must be in accordance with IE Rules or Rules framed under the Act.

2. Each Designated Officer shall maintain a legibly written safety logbook, in chronological order, of all operations and messages relating to safety co-ordination sent and received by him. All safety logs shall be retained for a period of not less than 10 years.

10. Operational Event/Accident Reporting.

- 10.1. Introduction -

This Section describes the reporting procedure of reportable events in the State Transmission System

This Section applies to SLDC, STU and all entities embedded within State power system that are under the control and supervision of SLDC.

10.2

Objective. - The objective of this Section is to define the events/ incidents to be reported, the reporting route to be followed and the information to be supplied to ensure a consistent approach in reporting of incidents and accidents in the State Transmission System.

10.3

Reportable Events. - Any of the following events that could affect the State Transmission System requires reporting:

- a. Exceptionally high / low system voltage or frequency.
- b. Serious equipment problem relating to major circuit breaker, transformer or bus bar.
- c. Loss of major Generating Unit. System split, State Transmission System breakaway or Black Start.
- d. Tripping of Transmission Line, ICT (Interconnecting transformer) and capacitor banks.
- e. Major fire incidents.
- f. Force-Majeure condition like flooding or lightning etc.
- g. Major failure of protection.
- h. Equipment and Transmission Line overload.
- i. Accidents-Fatal and Non-Fatal.
- j. Load Crash / Loss of Load
- k. Excessive Drawal deviations.
- l. Minor equipment alarms.

The last two reportable incidents are typical examples of those which are of lesser consequence, but which still affect the State Transmission System and can be reasonably classed as minor. They will require corrective action but may not warrant management reporting until these are repeated for sufficient time.

10.4

Reporting Procedure. - 1. Reporting Time for events and accidents. - All reportable incidents occurring on lines and equipment of 33 kV and above and all the lines on which there is the inter user flow affecting the State Transmission System shall promptly be reported orally by the User whose equipment has experienced the incident (the reporting User) to any other significantly affected Users and to SLDC. The reporting user should submit a written confirmation to

SLDC within one hour of such oral report.

If the reporting incident is of major nature then the Reporting User shall submit an initial written report within two hours to SLDC. This has to be further followed up by the submission of a comprehensive report within 48 hours of the submission of the initial written report. In other cases the Reporting User shall submit a report within 5 (five) days to SLDC.

2. SLDC shall call for a report from any User on any reportable incident affecting other Users and STU, in case the same is not reported by such User whose equipment might have been source of the reportable incident.

The above shall not relieve any User from the obligation to report events in accordance with IE Rules 1956 - (Language used in the Code to be used). The format of such a report shall be as agreed by the State Grid Code Review Panel, but will typically contain the following information:

- i. Location of incident.
- ii. Date and time of incident.
- iii. Plant or equipment involved.
- iv. Details of relay indications with nature of fault implications, Antecedent conditions like line flows, bus voltage, generation and demand
- v. Supplies and quantum interrupted and duration if applicable.
- vi. Amount of generation lost if applicable and its duration
- vii. Brief description of incident.
- viii. Estimate of time to return to service.
- ix. Name of Organisation
- x. Possibility of alternate arrangement of supply

10.5

Reporting Form. - The standard reporting Form other than for accidents shall be as agreed from time to time by the Grid Code Review Panel. A typical form is attached (APPENDIX-E).

10.6

Major Grid Incidence. - (a) Following a major grid incident, SLDC and other Users shall cooperate to enquire and establish the cause of such failure and make appropriate recommendations. SLDC shall report the occurrence of such major grid failure to the Commission in writing as well as ERLDC immediately for information and shall submit the enquiry report to the Commission within two months of the incident. Analysis of major grid disturbance in the Intra State Power System soon after their occurrence shall be done by a Protection subcommittee constituted by STU.

(b)

Periodic Reports - All distribution licensees shall send a weekly report to SLDC on the performance of their respective systems which should cover the following information:

(i)

Voltage profile at all Sub/Stations (66KV) 132KV and above.

(ii)

Average, maximum, minimum demand (both MW and MVAR) met at such S/Stns.

(iii)

Quantum and duration of load shed, with reasons

(iv)

Outage of major elements

(v)

Network constraints

(vi)

Daily energy consumed and energy exchanged by the DISCOM

(c)

SLDC shall post in its website a monthly performance report of the State as a whole covering:

(i)

Hourly demand met and generation for peak any and minimum demand met day. Also the average daily off-peak and peak demands met.

(ii)

Daily average consumption

(iii)

Station-wise daily maximum, minimum and average generation (MW), together with daily energy generation

(iv)

Instances of non-compliance of the State Grid Code

(v)

Progress of construction of new generating units, lines and transformers. Details of generation and transmission outages during the month.

10.7

Accident Reporting. - Report of accidents shall be in accordance with the Section 161 of the Electricity Act, 2003 and the Rules framed thereunder. Receipt of accident and failure of supply or transmission of electricity shall be in the specified Form to the Commission and the Electrical Inspector.

Chapter 4

Scheduling and Despatch Code

11. Scheduling and Despatch.

- 11.1. Introduction -

This Section specifies the procedure to be adopted for the scheduling and despatch of State Generating Station (SGS) and Inter-State Generating Station (ISGS) to meet system demand and drawal allocation requirements of Distribution Licensees.

11.2

Objective. - The objective of this Section is to detail the actions and responsibilities of SLDC in preparing and issuing a daily schedule of generation to the SGSs within its ambit, drawal schedule of various distribution licensees and other entities utilizing the Intra-State Transmission System, furnishing requisition from Eastern Region ISGS to ERLDC, including details of bilateral transactions to be scheduled with other regional entities.

11.3

General. - The following specific points would be taken into consideration while preparing and finalising the schedules:

1. SLDC will issue despatch instructions required to regulate all generation and imports from IPPs / CPPs according to the 15-minute day ahead generation schedule. In the absence of any despatch instruction by SLDC, SGS shall generate/ export according to the day ahead generation schedule.

2. The SLDC shall regulate the overall State generation in such a manner that generation from following types of power stations shall not be curtailed except under abnormal operating conditions.

(i)

Run of river or canal based hydro stations.

(ii)

Storage type hydro-stations when water level is at peak reservoir level or expected to touch peak reservoir level as per inflows.

(iii)

Generation from non-conventional sources like wind, solar that cannot be stored/ controlled.

3. Despatch instructions to SGS shall be in standard format to be finalized by SLDC.

4. The algebraic sum of scheduled drawals from ISGS, long term open access, and short term open access arrangements shall be the net drawal schedule for the control area of the utility.

11.4

Generation Scheduling. - (1) Steps in Scheduling. - Step by step procedure for scheduling of ISGS and SGS/IPP/CPP embedded in the State power system (not having the status of regional entity shall be as described below:

i. By 9.00 hours every day each SGS shall intimate to SLDC the station wise ex- power plant MW and MWh capabilities foreseen for the next day i.e. between 00.00 to 24.00 hrs of the following day, at 15 minutes interval.

ii. By 9.00 hours every day each Distribution Licensees bulk power consumer shall intimate SLDC the overall requirement in MW and MWh for the next day at 15 minutes interval.

iii. After receipt of the information in regard to the availability from different sources, the SLDC shall review aggregate generating capability of ISGS, SGS and the bilateral interchanges, if any, vis-a-vis Distribution Licensees requirements.

iv. By 15.00 hrs, SLDC shall finalise (a) generation schedule of SGS and (b) drawal schedule of each Distribution Licensees. It shall accordingly advise each Distribution Licensees of their drawal schedule and will workout and convey

to ERLDC for net drawal schedule in each of the ISGS along with the bilateral exchanges agreed or intended to be had with the other State/States and the estimates of demand / availability in the State and additional power it would like to draw subject to availability.

v. By 1700 hrs, ERLDC shall convey to SLDC the drawal schedule for Bihar State from each of the ISGS and other short-term bilateral, medium term and long-term schedules and SLDC shall convey to SGS the generation schedule and drawal schedule to Distribution Licensees by 1900 hrs.

vi. SGS and each Distribution Licensees may inform the modifications / changes to be made, if any, in the above schedule to SLDC by 21.30 hours.

vii. SLDC after considering the same shall convey revised schedule to ERLDC by 22.00 hrs.

viii. On receipt of information and after due consultations, the ERLDC shall issue the final generation and drawal schedule by 23.00 hrs, and SLDC shall inform the same to all concerned.

(2)

SLDC shall prepare the day ahead generation schedule keeping in view the following:

(i)

Transmission System constraints from time to time.

(ii)

15 minute load requirements as estimated by SLDC.

(iii)

The need to provide operating margins and reserves required to be maintained.

(iv)

The availability of generation from SGS, Central Sector Generators and others together with any constraint in each case.

(3)

During the day of operation, the generation schedule may be revised under following conditions:

i. In case of forced outage of a unit of any SGS, SLDC may revise the generation schedule on the basis of revised declared capability by the affected SGS.

ii. ERLDC may revise the schedule of drawal from Eastern Region and consequently SLDC shall enforce the revisions within Bihar.

11.5

Drawal Scheduling. - SLDC is responsible for collection, examination and compilation of drawal Schedule for each Distribution Licensee in prescribed manner and at the prescribed time. Each Distribution Licensee shall supply to SLDC 15- minute average demand estimates in MW for the day ahead.

11.6

Generation Despatch. - 1. SGS and embedded CPP/IPP shall comply promptly with a despatch instruction issued by SLDC unless this action would compromise the safety of plant or personnel. SGS and embedded CPP/IPP shall promptly inform SLDC in the event of any unforeseen difficulties in carrying out an instruction.

2. Despatch instructions shall be issued by E-Mail /Fax/ telephone, confirmed by exchange of name of operators sending and receiving the same and logging the same at each end. All such oral instructions shall be complied with forthwith and written confirmation shall be issued promptly by FAX, tele-printer or otherwise

11.7

Responsibilities. - 1. SLDC shall monitor actual power drawal against scheduled power drawal and regulate internal generation and demand to maintain this schedule. SLDC shall also monitor reactive power drawal and availability of capacitor banks.

2. Generating Stations within the State shall follow the despatch instructions issued by SLDC.

3. Distribution Licensees and Open Access Customers shall comply with the instructions of SLDC for managing load & reactive power drawal as per system requirement.

12. Frequency and Voltage Management.

- 12.1. Introduction -

This Section describes the method by which all Users of the State Transmission System shall co-operate with SLDC and STU in contributing towards effective control of the system frequency and managing the voltage of the State Transmission System. The State Transmission System normally operates in synchronism with the Eastern Region Grid

and ERLDC has the overall responsibility of the integrated operation of the Eastern Regional Power System. The constituents of the Region are required to follow the instructions of ERLDC for backing down generation, regulating loads, MVAR drawal etc. to meet the objective. SLDC shall accordingly instruct Generating Units to regulate Generation/Export and adhere to active and reactive power generation within their respective declared parameters. SLDC shall also regulate the load as may be necessary to meet the objective. The State Transmission System voltage levels can be affected by Regional operation. The STU/SLDC shall optimize voltage management by adjusting transformer taps (On Line Tap Changers) to the extent available and switching of circuits/ capacitors/ reactors and other operational steps. SLDC will instruct SGS to regulate MVAR generation within their declared parameters. SLDC shall also instruct Distribution Licensees to regulate demand, if necessary by operating reactors/ capacitor banks nearest to load point in 33KV system.

12.2

Objective. - The objectives of this Section are as follows:

(1)
To define the responsibilities of all Users in contributing to frequency and voltage management.

(2)
To define the actions required to enable SLDC and STU to maintain the State Transmission System voltages and frequency within acceptable levels in accordance with IEGC guidelines as well as Planning and Security Standards for the State Transmission System specified by the Commission, if any.

12.3

Frequency Management. - (1) The rated frequency of the system shall be 50 Hz and shall normally be regulated within the limits prescribed in IEGC Clause 4.6(b) as also specified in Connection Conditions. STU and SLDC shall make all possible efforts to ensure that grid frequency remains within 49.5 Hz- 50.2 Hz band.

(2)
Falling frequency. - Under falling frequency conditions, SLDC shall take appropriate action to issue instructions, in co-ordination with ERLDC to arrest the falling frequency and restore frequency within permissible range. Such instructions may include despatch instruction to SGS and/or instruction to Distribution Licensees and Open Access Customers to reduce load demand by appropriate manual and/or automatic load shedding.

(3)
Rising frequency. - Under rising frequency conditions, SLDC shall take appropriate action to issue instructions to SGS in co-ordination with ERLDC to arrest the rising frequency and restore frequency within permissible range. SLDC shall also issue instructions to Distribution Licensees and Open Access Customers in coordination with ERLDC to lift Load shedding (if exists) in order to take additional load. In case of Load Crash, SLDC shall take steps as per Clause 6.6 of the Grid Code.

12.4

Voltage Management. - (1) Users using the State Transmission System shall make all possible efforts to ensure that the grid voltage always remains within the limits specified in IEGC at clause 5.2 (r) and IE Rules 1956 as re-produced below:

Voltage (KV r ms)

Nominal

Maximum

Minimum

400

420

380

220

245

198

132

145

122

(2)

STU and/or SLDC shall carry out load flow studies based on operational data from time to time to predict where voltage problems may be encountered and to identify appropriate measures to ensure that voltages remain within the defined limits. On the basis of these studies, SLDC shall instruct SGS to maintain specified voltage level at interconnecting points.

SLDC shall continuously monitor 220/132kV voltage levels at strategic substations.

(3)

SLDC shall take appropriate measures to control State Transmission System voltages, which may include but not be limited to transformer tap changing, capacitor/reactor switching including capacitor switching by Distribution Licensees at 66 KV & 33 KV substations, operation of Hydro unit as synchronous condenser and use of MVar reserves with State Generating Stations within technical limits agreed to between STU and Generators. Generators shall inform SLDC of their reactive reserve capability promptly on request.

(4)

Distribution Licensees and Open Access Customers shall participate in voltage management by providing Local VAR compensation (as far as possible in low voltage system close to load points) such that they do not depend upon EHV grid for reactive support.

13. Monitoring of Generation and Drawal.

- 13.1. Introduction - The monitoring of SGS output, active and reactive reserve capacity is important to evaluate the performance of generation plants.

The monitoring of actual drawal against schedule is important to ensure that STU and Distribution Licensees contribute towards improving system performance and observe Grid discipline.

13.2

Objective. - The objective of this Section is to define the responsibilities of all SGS in monitoring of Generating Unit's reliability and performance, and STU's/ Discoms' compliance with the scheduled Drawal to assist SLDC in managing voltage and frequency.

13.3

Monitoring Procedure. - (1) For effective operation of the State Transmission System, it is important that a SGS's declared availability is realistic and that any departures are continually and invariably fed back to the Generator to help effect improvement.

(2)

The SLDC shall continuously monitor Generating Unit outputs and Bus voltages. More stringent monitoring may be performed at any time when there is reason to believe that a SGS's declared availability may not match the actual availability or declared output does not match the actual output.

(3)

SLDC can ask for putting a generating station to demonstrate the declared availability by instructing the generating station to come up to the declared availability within time specified by Generators.

(4)

SLDC shall inform SGS, in writing, if the continual monitoring demonstrates an apparent persistent or material mismatch between the despatch instructions and the Generating Unit output or breach of the Connection Conditions. Continued discrepancies shall be resolved by the Grid Code Review Panel with a view to either improve performance in future, providing more realistic declarations or initiate appropriate actions for any breach of Connectivity Conditions. Continued default by a generating station entails penalty as may be determined by the Commission.

(5)

SGS (excluding CPPs) shall provide to SLDC 15-minute block-wise generation summation outputs where no automatically transmitted metering or SCADA/RTU equipment exists. CPPs shall provide to SLDC 15-minute block-wise export / import MW and MVar.

(6)

The SGS shall provide any other logged readings that SLDC may reasonably require, for monitoring purposes where SCADA data is not available.

13.4

Generating Unit Trippings. - (1) SGS shall promptly inform SLDC of the tripping of a Generating Unit, with reasons in accordance with Section 10 'Operational Event/Accident Reporting'. SLDC shall intimate ERLDC about the trippings and

their restoration. SLDC shall keep a written log of all such trippings, including the reasons with a view to demonstrating the effect on system performance and identifying the need for remedial measures.

(2)
SGS shall submit a more detailed report of Generating Unit tripping to SLDC on monthly basis.

13.5
Monitoring of Drawal. - (1) SLDC shall continuously monitor actual MW Drawal by Distribution Licensees and other users against their schedules through use of SCADA equipment wherever available, or otherwise using available metering. SLDC shall request ERLDC and adjacent States as appropriate to provide any additional data required to enable this monitoring to be carried out.

(2)
SLDC shall continuously monitor the actual MVar drawal to the extent possible. This will be used to assist in State Transmission System voltage management.

13.6
Data Requirement. - SGS shall submit data to SLDC as listed in Data Registration Section (Appendix C-2)

Chapter 5 Protection Code

14. Protection. - 14.1. introduction -

In order to safeguard the State Transmission System and Users' system from faults occurring in other User's system, it is essential that certain minimum standards for protection be adopted. This Section describes these minimum standards.

14.2
Objective. - The objective of this Section is to define the minimum protection requirements for any equipment connected to the State Transmission System and thereby minimise disruption due to faults. This code applies to all users of the STU.

14.3
General Principles. - STU, SGS, other embedded generators DISCOMS and other bulk consumers shall abide by the provisions contained in Section 6, Part-1 of the CEA regulations on Technical Standards for connectivity to the grid.

(1)
No item of electrical equipment shall be allowed to remain connected to the State Transmission System unless it is covered by minimum specified protection aimed at reliability, selectivity, speed, stability and sensitivity.

(2)
All Users shall co-operate with STU to ensure correct and appropriate settings of protection to achieve effective, discriminatory removal of faulty equipment within the target clearance time specified in this Section.

(3)
Protective Relay settings shall not be altered, or protection relays bye-passed and/or disconnected without consultation and agreement between all affected Users. In a case where protection is bye-passed and/or disconnected by an agreement, then the cause must be rectified and the protection restored to normal condition as quickly as possible. If agreement has not been reached, the electrical equipment shall be removed from service forthwith.

14.4
Protection Coordination. - The settings of protective relays starting from the generating unit upto the remote end of 132 kV / 33 kV lines shall be such that only the faulty Section is isolated under all circumstances. The STU /Transmission Licensee shall notify the initial settings and any subsequent changes to the Users from time to time. Routine checks on the performance of the protective relays shall be conducted and any malfunction shall be noted and corrected as soon as possible. The STU / Transmission Licensee shall conduct the required studies for deciding the relay settings, with the data collected from the Users. If necessary, Protection Coordination Committee with representatives from the generating companies, STU and Distribution Licensees and bulk consumers connected to STU shall be constituted to meet periodically to discuss any malfunctions, changes in the system configuration, if any, and possible revised settings of relays.

14.5
Fault Clearance Times & Short-time Ratings. - (1) From stability consideration, the minimum short circuit current rating and time and the maximum fault clearance times for faults on any User's system directly connected to the State

Transmission System, or any faults on the State Transmission System itself, are as follows:

Nominal Voltage

Minimum Short Circuit current rating &

duration for Switchgear

Target Fault clearance Time

KV

KA(rms)

Seconds

msec.

220 KV

40

1

160

132 KV

40

1

160

(2)

Slower fault clearance times for faults on a Users system if it already exists, may be agreed to but only if, in STU's opinion, system conditions allow this. STU shall specify the required opening time and rupturing capacity of the circuit breakers at various locations for STU and Distribution Licensees / Open Access Customers directly connected to Transmission System. At generating stations, line faults should be cleared at the generation station end within the critical clearing time so that the generators remain in synchronism.

14.6

Generator Requirements. - All Generating Units and all associated electrical equipment of the Generating Units connected to the State Transmission System shall have adequate protection so that the State Transmission System does not suffer due to any disturbances originating from the Generation units. The generator protection schemes shall cover at least Differential protection, back up protection, Stator & Rotor Earth fault protection, field ground/field failure protection (not applicable to brush-less excitation system), negative sequence protection, under frequency, over flux protection, inter-turn Differential protection for generator, restricted E/F for Generator Transformer, back- up impedance protection, pole slipping protection (applicable to units above 200MW), reverse power protection etc.

14.7

Transmission Line Requirements. - (1) General. - Every EHV line taking off from a Generating Station or a sub-station or a switching station shall have protection and back up protection as mentioned below. STU. shall notify Users of any changes in its policy on protection. Switchgear equipment and Relay Panels for the protection of lines of STU taking off from a Generating Station shall be owned and maintained by the Generator. Any transmission line related relay settings or any change in relay settings will be carried out by the Generator in close co-ordination and consultation with STU.. Carrier cabinets / equipment, Line matching units including wave traps and communication cable shall be owned and maintained by STU. All Generators shall provide space, connection facility, and access to STU for such purpose.

(2)

220 KV Transmission Lines. - All 220 KV transmission lines owned by STU shall have two fast acting protection schemes.

Main 1 Protection scheme shall be numeric, three zone, non-switched fast acting distance protection scheme with permissible inter-trip at remote end (in case of zone-2 fault). The scheme shall have power swing blocking, location of fault recording, disturbance recording, event logger, communication port, as well as Local Breaker Backup (LBB).

Main 2 Protection scheme shall be preferably numeric, three zone, switched/ non-switched fast acting distance protection scheme having all features as main-1 except auto reclosing and Local Breaker Backup (LBB). This protection should preferably be fed from separate core of CT and DC source.

For back-up protection, three directional IDMTL over current relays and unidirectional earth fault relay shall be provided.

(3)

132 KV Lines. -

? A single scheme three zone, non-switched numeric distance protection with standard built in features like single and three phase tripping, carrier inter-tripping, IDMT over current and earth fault, power swing blocking and LBB protection shall be provided as main protection.

? The backup protection shall be at least two directional IDMTL over current relays and one directional earth fault relay.

? For short transmission radial lines, appropriate alternative protection schemes may be adopted.

? For 33 KV feeders, emanating from Grid S/S, adequate protection with proper co-ordination should be provided at Grid S/S.

14.8

Transformer Requirements. - (1) The protection of EHV Transformers, Power Transformers shall be as per revised manual on transformers published by Central Board of Irrigation and Power (CBIP) Publication No. 275.

The following minimum protections should be provided for transformers:

i. All 220 KV class power transformers shall be provided with numeric fast acting differential, REF, open delta (Neutral Displacement Relay) and over-fluxing relays. In addition, there shall be back up IDMTL over current and earth fault protection with directional features. For parallel operation, such back up protection shall have inter-tripping of both HV and LV breakers. For protection against heavy short circuits, the over current relays should incorporate a high set instantaneous element. In addition to electrical protection, transformer own protection viz. Buchholz, OLTC oil surge, gas operated relays, winding temperature protection, oil temperature protection, PRV relay shall be provided for alarm and trip functions.

ii. For 132 KV and 33 KV class transformers of capacity 10 MVA and above, the protection shall be same as mentioned in 14.8 (1) (i) except over-fluxing, REF and PRV relays.

iii. For 132 KV and 33 KV class power transformers less than or equal to 10 MVA provided on either Transmission or Distribution System, over current and E/F with high set instantaneous element along with auxiliary relays for transformer trip and alarm functions as per transformer requirements, shall be provided.

Transformer shall be protected from lightning arrestors from H.V and L.V both sides.

Pressure Release Valve (PRV) and differential protection should be installed for protection of 132 KV class Transformer of Capacity 10 MVA and above.

(2)

In addition to electrical protection, gas operated relays, winding temperature protection and oil temperature protection shall be provided in all transformers.

14.9

Sub-Station Fire Protection. - Adequate precautions shall be taken and protection shall be provided against fire hazards to all Apparatus of the Users conforming to relevant Indian Standard Specification and provisions in I.E. Rules 1956 or Rules framed under Electricity Act 2003.

14.10

Calibration and Testing. - The protection scheme shall be tested at each 220 KV, 132 KV substation by STU once in a year or immediately after any major fault, whichever is earlier.

Setting, co-ordination, testing and calibration of all protection schemes pertaining to generating units/stations shall be responsibility of respective SGC.

Chapter 6

Metering Code

15.

Metering Code.

- 15.1. Introduction -

This code prescribes a uniform policy in respect of electricity metering in the State Transmission System amongst the utilities i.e. STU, Generating Companies, Distribution Licensees and for the Open Access Customers on the State Transmission system and EHV Consumers (of Distribution Licensees) directly connected to the State Transmission System.

15.2

Objective. - The objective of this Section is to define minimum acceptable standards of metering which shall provide proper metering of various operating system parameters for the purpose of accounting, commercial billing and settlement of electrical energy and to provide information which shall enable to operate the system in economic manner.

15.3

Scope. - (1) The scope of this code covers the practices that shall be employed and the facilities that shall be provided for the measurement and recording of various parameters like active/reactive/apparent power/energy, power factor, voltage, frequency etc.

(2)

This code sets out or refers to the requirements of metering at generating stations, sub-stations and interfaces for tariff and operational metering.

(3)

This code also specifies the requirement for calibration, testing and commissioning of metering equipments viz. energy meters with associated accessories, current transformers and voltage transformers. The code broadly indicates the technical features of various elements of the metering, data communication and testing system.

15.4

Applicability. - This Metering Code shall apply to:

(ii)

STU/Transmission Licensees

(iii)

Generating Stations connected to State Transmission System

(iv)

Distribution Licensees connected with State Transmission System

(v)

EHV Consumers of Distribution Licensee(s) directly connected to State Transmission System

(vi)

Open Access Customers availing Open Access on State Transmission system

(vii)

Captive Generators connected to State Transmission System

15.5

Reference Standards. - All the equipment installed under this Code shall necessarily conform to the relevant standards as specified in the Central Electricity Authority's Standards/Regulations on Installation and Operation of Meters notified on 17/03/2007

15.6

Meter Installation. - 1. Ownership. - The ownership of the metering system shall be as provided in relevant agreement governing exchange of power and if no agreement exists then the ownership of the metering system shall belong to the User in whose premises the metering equipment is installed.

2. Right to Install Energy Meters. - Each User shall extend necessary assistance and make available the required space to the other User for installation of the metering equipment and provide required outputs of the specified current and voltage transformers to facilitate installation of Meters, RTUs and associated equipment in their premises.

3. Access to Equipment and Data. - Each User on request, shall grant full right to install metering equipments and RTUs to other User's employees, agents/duly authorized representative. The other Users shall also have access to metering locations for inspecting, testing, calibrating, sealing, replacing the damaged equipment, collecting the data, joint readings of meters and metering equipments, and other functions necessary jointly or otherwise as mutually agreed.

4. Operation and Maintenance of the Metering System. - The operation and maintenance of the metering system includes proper installation, regular maintenance of the metering system and RTUs, checking of errors of the CTs, VTs and meters, proper laying of cables and protection thereof, cleaning of connections/joints, checking of voltage drop in the CT/VT leads, condition of meter box and enclosure, condition of seals, regular/daily reading meters and regular data retrieved through CMRI, attending any breakdown/fault on the metering system etc.

5. Type of Meters and Metering Capability. - The meters shall be all electronic (static) poly phase tri-vector type having facility to measure active, reactive and apparent energy/power in all four quadrants i.e. a true import-export meter. All inter user meters shall be bi-directional while capacitor bank meters and substation aux. meters may be unidirectional. ABT compliant energy meters shall be provided at such interface points, wherever the energy exchange is based on Availability Based Tariff (ABT).

15.7

Various Standards for Metering Equipment. - (1) The minimum specifications for the metering equipment are given below.

Sl. No.

Particulars

Meter Type

Main & Check

Back up

Capacitor Bank

Sub-Station Auxiliary

Inter Distribution Licensees

Secondary Back up

1

2

3

4

5

6

7

8

(1)

Accuracy class

(a) Meter

0.2 S

0.2 S

0.5 S

1.0

0.2 S

0.2 S

(b) CTs

0.2

0.2

0.2

0.2

0.2

0.2

(c) PTs / CVTs

0.2

0.2

0.2

0.2

0.2

0.2

(d) CT-PT sets for 33 KV & 11 KV feeders

0.2

Existing

Existing

Existing

Existing

Existing

(2)

Salient aspect of meters

(a) Phase angle and ratio error compensation of
CTs & PTs

No

No

No

No

No

No

(b) Communication port

(i) Optical port

Yes

Yes

No

No

Yes

Yes

(ii) For remote reading

Yes

Yes

No

No

Yes

Yes

(c) Whether both Import & Export features
required

Yes

Yes

No

No

Yes

Yes

(d) Meter memory for 45 days

Yes

Yes

No

No

Yes

Yes

(2)

Minimum Technical Requirements for Energy Meter

i. Operating System Parameters (for balanced and unbalanced load):

a. Operating Voltage Range: The meter shall work satisfactorily on 110 Volts AC (Line-Line) or 415 Volts AC (Line-Line) with variation range of -40% to +20%.

b. Operating Frequency Range: The meter shall work satisfactorily on 50 Hertz with variation range of -5% to +5%.

c. Operating Power Factor Range: The meter shall work satisfactorily over a power factor range of zero lag to unity to zero lead.

ii. Measuring Elements:

a. The meter shall be 3 phase 4 wire type, capable to record and display import and export kWh, kVAh, kVAh and maximum demand in kW and kVA for 3 phase 4 wire AC balanced/unbalanced load for a power factor having range of zero lagging to unity to zero leading in all 4 quadrants. In addition, meter shall also be capable of displaying, on demand, the present status of supply/load, missing potential, CT polarity, current unbalance, anomaly occurrence and logging of

occurrences as well as load survey data etc. which shall be down loaded to a user friendly Base Computer System (BCS) through portable data collection devices or CMRI which shall be connected to optical communication port of the meter. Meter shall be equipped with self-diagnostic features also and be capable of recording average values based on their integration on time base for kWh, kVAh, kVAh for at least 45 days. Meter shall be capable of measuring fundamental as well as total energy including harmonics separately.

b. Energy measurement during demand period shall be such that sampling in the meter is synchronized with the end of the time block otherwise energy measured in a demand period but not stored in that period shall be carried forward. An LED glow or pulse output coincident with end of each demand period need be provided in the meter so as to ensure that demand integration coincided the preset time block.

iii. Display. - Present meter status, real time and date, cumulative energy registers, voltage, currents, power factor, present demand, frequency and meter serial number shall be available on demand through push button. Any interrogation/read operation shall not delete or alter any stored meter data.

iv. Memory. - Numerical values of voltage/current, power factor and cumulative energy registers as well as anomalies/tampered details alongwith date and time of logging of and restoration of anomalies (subject to the meter memory space) shall be logged in the meter memory and shall be available for retrieving with the help of the data collection devices (CMRI) through meter optical port and down loading to BCS.

a. Memory in a static tri-vector meter shall not get 'erased' after reading or retrieving of data through CMRI. Data shall be retained for a minimum of 45 days or shall not get erased from meter until replaced by fresh data. However, desired data can be erased from CMRI, when memory of a CMRI becomes full after downloading of readings of a number of meters, as there is fixed space made available in CMRI for:

- i. Energy registers.
- ii. Load survey data.
- iii. Anomaly data etc.

When a fresh data is logged in the memory, the oldest data shall disappear automatically.

v. Test terminal blocks . - The test terminal blocks shall be provided on all meters to facilitate testing of meters in service. Main & back up meters of inter state / major generating stations shall be having the feature of draw out type modular units and shall have automatic CT short circuiting so that meter can be taken out for testing without shut down requirements.

vi. Meter Power Supply. - Meters of inter state / major generating stations shall be capable of powered with 230 volt alternating current auxiliary supply and 110 volt or 220 volt DC supply of the substation so that metering core of PT/ CVT is never loaded and in case of shut down on feeder/breaker, meter can be interrogated locally or remotely. It shall normally be powered by AC auxiliary supply and shall be switched over to DC supply only when AC auxiliary supply fails.

vii. Battery back-up . - The meter shall have battery back up for its Real Time Clock (RTC).

viii. Meter Programmability. - The meters shall be equipped with necessary hardware/software to suit tariff requirements such as ABT, TOD, two-part tariff based on SMD as may be called for from time to time.

(3)

Minimum Technical Requirement for Current Transformer (CT)

a. Three single-phase type current transformers shall be used for 3 phase 4 wire and 3 phase 3 wire measurement system. The secondary current rating of the CTs shall be 1 Ampere particularly for 220 KV, 132 KV, 66 KV measurements. However existing CTs with 5-Ampere secondary current shall also be acceptable provided the connected meters and instrument have base 5 Ampere current rating. For other voltages, 1 Ampere or 5 Ampere shall be employed.

b. The current transformers shall have dedicated core for metering and wherever feasible, the cores feeding to main meters and check meters shall be separate. The errors of the current transformers shall be checked in the laboratory or at site. However if such facilities are not available, CT test certificates issued by a Government test house or Government recognized test agency shall be referred to.

c. The total burden connected to each current transformer shall not exceed the rated burden of CT. Total circuit burden shall be kept close to rated burden of CT for minimum error.

(4)

Minimum Technical Requirement for Voltage Transformers (VT)

- a. Either Electromagnetic Voltage Transformers (EVT) or Capacitive Voltage transformer (CVT) should be used for metering purpose. Generally, term VT is used to cover either EVT or CVT. The secondary voltage per phase shall be 110/31/2 volts or 415/31/2 volts. Either dedicated VTs or dedicated core of VTs shall be provided for metering and that wherever feasible, VTs (or their cores) feeding to main meters and backup/check meters shall be separate. Fuses of proper rating shall be provided at appropriate locations in the VT circuit,
- b. The errors of the VTs shall be checked in the lab or at site. However if such facilities are not available, VT test certificates issued by Government test house or Government recognized test agency should be referred to.
- c. The total burden connected to each VT shall not exceed the rated burden of VT. Voltage drop in VT leads shall be within the permissible limits.
- d. The current transformers and voltage transformers shall meet the requirements as per the relevant standards. Where a combined CT/PT unit is provided, the accuracy shall be as specified under relevant IS.

15.8

Testing Arrangement. - (1) Two types of test facilities shall be available:

- a. Meter test bench with high accuracy, static source and 0.02S class electronic reference standard meter (RS Meter) shall be used for testing and calibration of meters. Meter Testing Laboratories duly equipped with testing benches and other equipments shall be established at suitable locations for testing and calibration of meters by SGS, STU and Distribution Licensee. The Meter Testing benches with 0.02S-class reference standard meter shall also be used for checking and calibration of portable testing equipments. Testing, calibration and maintenance of Energy Meters shall conform to the requirement of IS: 9792 and Testing equipments shall conform to Indian Standards Specification IS: 12346.
- b. Portable test set with static source and electronic reference meter of 0.1 class shall be used for verification and joint testing of accuracy of static tri-vector meters at site on regular/routine basis.

(2)

Separate test terminal blocks for testing of main and check meters shall be provided so that when one meter is under testing, the other meter continues to record actual energy during testing period. Where only one/main meter exists, an additional meter shall be put in circuit to record energy during the testing period of the main meter so that while the main meter is under testing, the other meter continues to record energy during the period of meter remaining under testing.

(3)

Testing at site shall be carried out for all meters once in a year.

(4)

The Licensee shall allow the testing of Open Access Customers' meters at third party (NABL approved) Testing Labs in case the Customers so request for the same. In case of testing by third party (NABL approved) Testing Labs, the Open Access Customers shall apply with prescribed fee to the Licensee.

15.9

Meter Reading. - The STU and concerned Generating Companies, CPP /Distribution Licensees, Open Access Customers as the case may be shall jointly read the meters through their authorized representatives preferably on 1st of every month at 12.00 Hrs. / retrieve meter reading data using CMRV Tele metering.

15.10

Joint Inspection, Testing, Calibrations. - (1) The metering system located at metering points between Generating Companies, STU and Distribution Licensees shall be regularly inspected at least once in a year or at an interval lesser than 1 year as mutually agreed by both the agencies involved for despatch and receipt of energy. Since the static tri-vector meters are calibrated through software at the manufacturers' works, only accuracy of the meters and functioning shall be verified during joint inspection and certified jointly by both the agencies. After testing, the meter shall be properly sealed and a joint report shall be prepared giving details of testing work carried out, details of old seals removed and new seals affixed, test results, further action to be taken (if any) etc. The agency in whose premises the meter is located shall be responsible for proper security and protection of the metering equipment and sealing arrangement.

(2)

Joint inspection shall also be carried out as and when difference in meter readings exceeds the sum of maximum error as per accuracy class of main and check meter. The meters provided at the sending end as well as at the receiving end shall be jointly tested/ calibrated on all loads and power factors as per relevant standards through static phantom load.

15.11

Sealing. - i. Tariff metering systems shall be jointly sealed by the authorized representatives of the concerned agencies as per the procedure agreed upon.

ii. Any seal, applied, shall not be broken or removed except in the presence of or with the prior consent of the agency affixing the seal or on whose behalf the seal has been affixed unless it is necessary to do so in circumstances where (a) both main and check meters are malfunctioning or there occurs a fire or similar hazard and such removal is essential and such consent can not be obtained immediately (b) such action is required for the purpose of attending to the meter failure. In such circumstances, verbal consent shall be given immediately and it must be confirmed in writing forthwith.

iii. Each agency shall control the issue of its own seals and sealing pliers, and shall keep proper register/record of all such pliers and the authorized persons to whom these are issued.

iv. Sealing of the metering system shall be carried out in such a manner so as not to hamper downloading of the data from the meter using CMRI or a remote meter reading system.

15.12

Interface Metering Arrangement. - The metering system shall comprise of main, check, backup and secondary backup meters. In the event of main meter becoming defective the order of precedence for billing shall be (a) main (b) check (c) backup (d) secondary backup.

Generating Stations:

a. Meters shall be installed on each Generator terminal, at each Unit Auxiliary Transformer (UAT), and all outgoing feeders at Generating Stations to work out energy generated and net energy delivered by the Power Station in the Grid.

b. For measurement of energy supplied by major generating stations within the state, meters shall be provided on each outgoing feeder at the power station designated as main meter for billing purpose as per commercial agreement and/or Grid Code Connectivity Conditions.

c. A Check Meter shall also be provided along with the Main Meter. Meters on each generator and each auxiliary transformer shall work as backup meters.

(2)

Interstate Transmission and Inter-Regional Transmission System. - Metering arrangement for Inter-State Transmission Lines and for Inter- Regional Transmission System shall be governed by IEGC. Special Energy Meters (SEM) capable of time-differentiated measurement (15 minutes) of active energy and voltage differentiated measurement of reactive energy as specified by CTU/ERLDC shall be provided on interstate and inter-regional transmission lines. STU shall comply with requirement for installation, meter reading & downloading and communication of readings of Special Energy Meters (SEM) to ERLDC as per operating procedure of ERLDC. STU may install its own Check Meters at inter-state/inter-regional transmission lines at the periphery of State Transmission System.

(3)

Metering between STU-Distribution Licensee

a For measurement of power delivered by STU to Distribution Licensee, metering shall be provided on the LV side of EHV Power Transformer i.e. 33 KV side of 220/33 KV and 33 KV side of 132/33 KV and 11 KV side of 132/33/11 KV and 11 KV side of 132/11 KV transformers installed in EHV sub-stations.

b Operational meters shall also be provided on all outgoing 33 KV and 11 KV feeders as back-up meter for energy audit on feeder and reconciliation of energy with respect to energy measured on LV side of EHV Power Transformer.

c. In case of EHV industrial and other consumers directly fed from 220 KV or 132 KV sub-stations, tariff metering shall be provided on outgoing feeder emanating from EHV sub-station.

(4)

Metering between two Distribution Licensee:

a. The energy metering shall be provided at such points of the power lines connecting any two Distribution Systems owned by different Distribution Licensees so that the measured energy gives correct measurement of consumption by either Distribution Licensee.

b. If installation of metering at such point is not feasible, it shall be provided at nearest sub-station feeding other Distribution System. In such case, energy accounting may be in proportion to installed capacity of Distribution Transformers on the line or as agreed mutually.

(5)

Sub-station Auxiliary Consumption Metering. - The STU sub-stations auxiliary consumption shall be recorded on LV

side of station auxiliary transformers. If such transformer(s) is feeding other local load (colony quarters, streetlights etc.) apart from substation auxiliary load, separate metering shall be provided on feeder feeding the colony quarters, street lights, etc.

(6)

Open Access Customers. - The Inter-State Open Access Customers shall provide Special Energy Meters. The embedded Open Access Customers within the State Transmission System shall also provide Special Energy Meters both at the point of injection and point of drawal of supply. Special Energy Meters (SEM) shall be capable of time-differentiated measurement (15 minutes) of active energy and voltage differentiated measurement of reactive energy as specified by CTU/ RLDC. The Distribution licensee may provide Check Meters of the same specification as Main Meters.

(7)

Operational Metering. - Operational metering shall be provided wherever reasonably required by STU/ Generating Companies for applications other than tariff metering.

15.13

Supervisory Control and Data Acquisition (SCADA). - (1)The STU shall install and make operative an operational Metering Data Collection System under SCADA for storage, display and processing of Operational Metering Data. All Users shall make available outputs of their respective Operational meters to the SCADA interface equipment.

(2)

The data collection, storage and display centre shall be the State Load Despatch Centre (SLDC).

15.14

ABT, Two part and TOD Tariff Capability. - The metering arrangement for recording Distribution Licensee consumption/power input in his area of supply shall consist of following:

i. Frequency based ABT compliant meters shall be provided on 33 kV or lower voltage lines feeding each Distribution Licensee area of supply. The function of these meters will be as under:

a. To measure Distribution Licensee-wise UI (Unscheduled Interchange) energy and corresponding average frequency during 15 minute block.

b. The Distribution Licensee wise summation of kWh, kW, PF, demand, scheduled interchange/ unscheduled interchange will be done at the main computer station provided at central billing station or at Load Despatch Centre.

c. For this purpose, the various parameters shall be integrated at one centrally located station preferably at State Load Despatch Centre at Patna through computer and suitable software.

ii. Static tri-vector meters to be provided on LV secondary side of all EHV transformers. The function/duty of this meter will be as under:

a. Measurement of kWh energy supplied to Distribution Licensee for billing purpose.

b. kW/kVA demand and power factor, 15 minute block-wise as well monthly caused by Discom on each EHV transformer.

Chapter 7

Data Registration Code

16.

Data Registration.

- 16.1. Introduction -

This Section specifies a list of all data required by STU and SLDC, which is to be provided by the Users, and data required by Users to be provided by STU at times specified in the Grid Code. Other Sections of the Grid Code contain the obligation to submit the data and define the times when data is to be supplied by the Users.

16.2

Objective. - The objective of this Section is to list out all the data required to be provided by Users to STU and vice versa, in accordance with the provisions of the Grid Code.

16.3

Responsibility. - 1. All Users are responsible for submitting up-to-date data to STU/ SLDC in accordance with the provisions of the Grid Code.

2. All Users shall provide STU and SLDC with the name, address and telephone number of the person responsible for sending the data.

3. STU shall inform all Users and SLDC of the name, address and telephone number of the person responsible for receiving data.

4. STU shall provide up-to-date data to Users as provided in the relevant Sections of the Grid Code.

5. Responsibility for the correctness of data rests with the concerned User providing the data.

16.4

Data to be registered. - 1. Data required to be exchanged has been listed in the Appendices to this Section under various categories.

2. Changes to Users Data. - Whenever any User becomes aware of a change to any items of data that is registered with STU, the User must promptly notify STU of the changes. STU on receipt of intimation of the changes shall promptly correct the database accordingly. This shall also apply to any data compiled by STU regarding its own system.

3. Methods of Submitting Data. - a. The data shall be furnished in the standard formats for data submission and such formats must be used for the written submission of data to SLDC and STU.

Where standard formats are not appended they would be developed by SLDC or STU in consultation with Users.

b. All data to be submitted under the Schedule(s) must be submitted to SLDC / STU or to such other department and/or entity as STU may from time to time notify to Users. The name of the person who is submitting each schedule of data shall be indicated.

c. Where a computer data link exists between a User and SLDC/ STU, data may be submitted via this link. The data shall be in the same format as specified for paper transmission except for electronic encoding for which some other format may be more suited. The User shall specify the method to be used in consultation with the SLDC/ STU and resolve issues such as protocols, transmission speeds etc. at the time of transmission.

4. Data not supplied. - All Users are obliged to supply data as referred to in the individual Sections of the State Grid Code and listed out in the Data Registration Section Appendices. In case any data is not supplied by any User or is not available, STU or SLDC may, acting reasonably, if and when necessary, estimate such data depending upon the urgency of the situation. Similarly, in case any data is not supplied by STU, the concerned User may, acting reasonably, if and when necessary, estimate such data depending upon urgency of the situation. Such estimates will in each case, be based upon corresponding data for similar Plant or Apparatus or upon such other information, the User or STU or SLDC, as the case may be, deems appropriate.

16.5

Special Considerations. - STU and SLDC and any other User may at any time make reasonable request for extra data as necessary. STU shall supply data, required/requested by SLDC for system operation, from data bank to SLDC.

Appendix A

Standard Planning Data

A-1 Standard Planning Data (Generation)

For Sgs-Thermal

A. 1.1 Thermal (Coal / Gas/Fuel Linked)

(1)

General

i

Site

Furnish location map to scale showing roads, railway lines, Transmission lines, canals, pondage and reservoirs if any.

ii

Coal linkage/ Fuel (Like Liquefied Natural Gas, Naphtha etc.) linkage

Give information on means of coal transport/carriage. In case of other fuels, give details of source of fuel and their transport.

iii

Water Sources

Give information on availability of water for

operation of the Power Station.

iv

Environmental

State whether forest or other land areas are affected.

v

Site Map (To Scale)

Showing area required for Power Station coal linkage, coal yard, water pipe lines, ash disposal area, colony etc.

vi

Approximate period of construction

(2) Connection

i

Point of Connection

Furnish single line diagram of the proposed Connection with the system.

ii

Step up voltage for Connection (kV)

(3) Station Capacity

i

Total Generating Station capacity (MW)

State whether development will be carried out in phases and if so, furnish details.

ii

No. of units & unit size (MW)

(4) Generating Unit Data

i

Steam Generating Unit

State type, capacity, steam pressure, steam temperature etc.

ii

Steam turbine

State type, capacity.

iii

Generator

Type

Rating (MVA)

Speed (RPM)

Terminal voltage (KV)

Rated Power Factor

Reactive Power

Capability

(MVA_r) in the range

0.95 of leading and 0.85 lagging Short Circuit Ratio

Direct axis

(saturated) transient reactance (% on MVA rating)

Direct axis

(saturated) sub-transient reactance (% on MVA rating)

Auxiliary Power

Requirement

MW and MVA_r

Capability curve

Ramp-up and ramp-down

rate

Generator Characteristic curve

iv

Generator

Transformer

Type

Rated capacity (MVA)

Voltage Ratio (HV/LV)

Tap change Range (+ %

to - %)

Percentage Impedance (Positive Sequence at Full

load)

A.1.2 Hydro Electric (For SGS)

(1)

General

i

Site

Give location map to scale showing roads,
railway lines, and transmission lines.

ii

Site map (To scale)

Showing proposed canal, reservoir area, water
conductor system, fore-bay, power house etc.

iii

Submerged Area

Give information on area submerged, villages
submerged, submerged forest land, agricultural land etc

iv

Whether storage type or run of river type

v

Whether catchment receiving discharges from
other reservoir or power plant.

vi

Full reservoir level

vii

Minimum draw down level.

viii

Tail race level

ix

Design Head

x

Reservoir level v/s energy potential curve

xi

Restraint, if any, in water discharges

xii

Approximate period of construction.

(2) Connection

i

Point of Connection

Give single line diagram proposed Connection with the Transmission System.

ii

Step up voltage for Connection (kV)

(3) Station Capacity

i

Total Power Station capacity (MW)

State whether development is carried out in phases and if so furnish details.

ii

No. of units & unit size (MW)

(4) Generating Unit Data

i

Operating Head (in Metres)

a. Maximum

b. Minimum

c. Average

Hydro Unit

Capability to operate as synchronous condenser

Water head versus discharges curve (at full and part load) Power requirement or water discharge while operating as synchronous condenser

i

Turbine

State Type and capacity

iii

Generator

Type

Rating (MVA)

Speed (RPM)

Terminal voltage (KV)

Rated Power Factor

Reactive Power

Capability (MVA_r) in the range 0.95 of leading and 0.85 of lagging MW & MVA_r capability curve of generating unit

Short Circuit Ratio

Direct axis transient

(saturated) reactance (% on rated MVA)

Direct axis

sub-transient (saturated) reactance (% on rated MVA)

Auxiliary Power Requirement (MW)

iv

Generator-Transformer

a.Type

b. Rated Capacity

(MVA)

c. Voltage Ratio

HV/LV

d. Tap change Range

(+% to -%)

e. Percentage Impedance (Positive Sequence at Full Load).

A.2 Standard Planning Data (Transmission)

For STU and Transmission Licensees

STU shall make arrangements for getting the required data from different Departments of STU/other transmission licensees (if any) to update its Standard Planning Data in the format given below:

- i. Name of line (Indicating Power Stations and substations to be connected).
- ii. Voltage of line (KV).
- iii. No. of circuits.
- iv. Route length (km).
- v. Conductor sizes.
- vi. Line parameters (PU values).
 - (a) Resistance/km
 - (b) Inductance/km
 - (c) Susceptance/ km
- vii. Approximate power flow expected- MW & MVAR.
- viii. Terrain of the route- Give information regarding nature of terrain i.e. forest land, fallow land, agricultural and river basin, hill slope etc.
- ix. Route map (to scale) - Furnish topographical map showing the proposed route showing existing power lines and telecommunication lines.
- x. Purpose of Connection- Reference to Scheme, wheeling to other States etc.
- xi. Approximate period of Construction.

A.3. Standard Planning Data (Distribution)

For distribution licensees

(1) General

i

Area Map (to scale) Furnish map of Bihar duly marked with the area of supply relevant for the Distribution Licence.

ii

Consumer Data Furnish categories of consumers, their numbers and connected loads.

iii

Reference to Electrical Divisions presently in charge of the Distribution.

(2) Connection

i

Points of Connection Furnish single line diagram showing points of Connection

ii

Voltage of supply at points of Connection

iii

Names of Grid Sub-Station feeding the points of Connection

(3) Lines and Substations

i

Line Data Furnish lengths of line and voltages within the Area. '

ii

Sub-station Data Furnish details of 132/11 KV sub-stations, 33/11 KV sub-station, capacitor installations

(4) Loads

i

Loads drawn at points of Connection.

ii

Details of loads fed at EHV, if any. Give name of consumer, voltage of supply, contract demand/load and name of Grid Substation from which line is drawn, length of EHV line from Grid Sub-station to consumer's premises.

iii

Reactive Power compensation installed

(5) Demand Data (for All Loads 1 Mw and Above)

i

Type of load State whether furnace loads, rolling mills, traction loads, other industrial loads, pumping loads etc.

ii

Rated voltage and phase

iii

Electrical loading of equipment State number and size of motors, types of drive and control arrangements.

iv

Sensitivity of load to voltage and frequency of supply.

v

Maximum Harmonic content of load.

vi

Average and maximum phase unbalance of load.

vii

Nearest sub-station from which load is to be fed.

viii

Location map to scale Showing location of load with reference to lines and sub-stations in the vicinity.

(6) Load Forecast Data

i

Peak load and energy forecast for each category of loads for each of the succeeding 5 years.

ii

Details of methodology and assumptions on which forecasts are based.

iii

Details of loads 1 MW and above.

a. Name of

prospective consumer.

b. Location and

nature of load.

c. Sub-Station from

which to be fed.

d. Voltage of supply.

e. Phasing of load.

Appendix B

Detailed Planning Data

B.1 Detailed Planning Data (Generation)

B.1.1

Thermal Power Stations (For SGS)

(1)

General

i. Name of Power Station.

ii. Number and capacity of Generating Units (MVA).

iii. Ratings of all major equipments (Boilers and major accessories, Turbines, Alternators, Generator Unit Transformers etc).

iv. Single line Diagram of Power Station and switch-yard.

v. Relaying and metering diagram.

vi. Neutral Grounding of Generating Units.

vii. Excitation control- (What type is used? e.g. Thyristor, Fast Brushless Excitors)

viii. Earthing arrangements with earth resistance values.

(2)

Protection and Metering

i. Full description including settings for all relays and protection systems installed on the Generating Unit, Generator unit Transformer, Auxiliary Transformer and electrical motor of major equipments listed, but not limited to, under Sec. 3 (General).

ii. Full description including settings for all relays installed on all outgoing feeders from Power Station switchyard, Tie circuit breakers, and incoming circuit breakers.

iii. Full description of inter-tripping of circuit breakers at the point or points of Connection with the Transmission System.

iv. Most probable fault clearance time for electrical faults on the User's System.

v. Full description of operational and commercial metering schemes.

(3)

Switch-Yard

i. In relation to interconnecting transformers:

1. Rated MVA.

2. Voltage Ratio.

3. Vector Group.

4. Positive sequence reactance for maximum, minimum, normal Tap. (% on MVA).

5. Positive sequence resistance for maximum, minimum, normal Tap. (% on MVA).

6. Zero sequence reactance (% on MVA).

7. Tap changer Range (+% to -%) and steps.

8. Type of Tap changer, (off/on load).

ii. In relation to switchgear including circuit breakers, isolators on all circuits connected to the points of Connection:

1. Rated voltage (kV).

2. Type of circuit breaker (MOCB/ABCB/SF6).

3. Rated short circuit breaking current (kA) 3 phase.

4. Rated short circuit breaking current (kA) 1 phase.

5. Rated short circuit making current (kA) 3 phase.

6. Rated short circuit making current (kA) 1 -phase.

7. Provisions of auto reclosing with details.

iii. In relation to the Lightning Arresters -Technical data

iv. In relation to the Communication - Details of communication equipment installed at points of connections.

v. In relation to the Basic Insulation Level (kV) -

1. Busbar.

2. Switchgear.

3. Transformer bushings.

4. Transformer windings.

(4)

Parameters of Generating Units

i. Rated terminal voltage (kV).

ii. Rated MVA.

iii. Rated MW.

iv. Speed (rpm) or number of poles.

v. Inertia constant H (MW Sec./MVA).

vi. Short circuit ratio.

vii. Direct axis synchronous reactance (% on MVA) X_d .

viii. Direct axis (saturated) transient reactance (% on MVA) X_d' .

ix. Direct axis (saturated) sub-transient reactance (% on MVA) X_d'' .

x. Quadrature axis synchronous reactance (% on MVA) X_q .

xi. Quadrature axis (saturated) transient reactance (% on MVA) X_q' .

xii. Quadrature axis (saturated) sub-transient reactance (% on MVA) X_q'' .

xiii. Direct axis transient open circuit time constant (Sec) $T'do$.

xiv. Direct axis sub-transient open circuit time constant (Sec) $T''do$.

xv. Quadrature axis transient open circuit time constant (Sec) $T'qo$.

xvi. Quadrature axis sub-transient open circuit time constant (Sec) $T''qo$.

xvii. Stator Resistance (Ohm) R_a .

xviii. Neutral grounding details.

xix. Stator leakage reactance (Ohm) X_l .

xx. Stator time constant (Sec).

xxi. Rated Field current (A).

xxii. Open Circuit saturation characteristic for various terminal Voltages giving the compounding current to achieve the same.

xxiii. MW and MVA_r Capability curve

(5)

Parameters of Excitation Control System

i. Type of Excitation.

ii. Maximum Field Voltage.

iii. Minimum Field Voltage.

iv. Rated Field Voltage.

v. Details of excitation loop in block diagrams showing transfer functions of individual elements using I.E.E.E. symbols.

vi. Dynamic characteristics of over - excitation limiter.

vii. Dynamic characteristics of under-excitation limiter.

(6)

Parameters of Governor

i. Governor average gain (MW/Hz).

ii. Speeder motor setting range.

iii. Time constant of steam or fuel Governor valve.

iv. Governor valve opening limits.

v. Governor valve rate limits.

vi. Time constant of Turbine.

- vii. Governor block diagram showing transfer functions of individual elements using I.E.E.E. symbols.
(7)

Operational Parameters

Minimum notice required to synchronize a Generating Unit from desynchronization.

- i. Minimum time between synchronizing different Generating Units in a Power Station.
- ii. The minimum block load requirements on synchronizing.
- iii. Time required for synchronizing a Generating Unit for the following conditions:
 1. Hot
 2. Warm
 3. Cold
- iv. Maximum Generating Unit loading rates for the following conditions:
 1. Hot
 2. Warm
 3. Cold
- v. (v) Minimum load without oil support (MW).

(8)

General Status

- i. Detailed Project report.
- ii. Status Report
 1. Land
 2. Coal
 3. Water
 4. Environmental clearance
 5. Rehabilitation of displaced persons
- iii. Techno-economic approval by Central Electricity Authority (CEA).
- iv. Approval of State Government/Government of India.
- v. Financial Tie-up.

(9)

Connection

- i. Reports of Studies for parallel operation with the State Transmission System.
- ii. Short Circuit studies
- iii. Stability Studies.
- iv. Load Flow Studies.
- v. Proposed Connection with the State Transmission System.
 - a. Voltage
 - b. No. of circuits
 - c. Point of Connection.

B.1.2 Hydro-Electric Stations (For SGS)

(1)

General

- i. Name of Power Station.
- ii. No and capacity of units. (MVA)
- iii. Ratings of all major equipment.
 - a. Turbines (HP)
 - b. Generators (MVA)
 - c. Generator Transformers (MVA)
 - d. Auxiliary Transformers (MVA)
- iv. Single line diagram of Power Station and switch-yard.
- v. Relaying and metering diagram.
- vi. Neutral grounding of Generator.
- vii. Excitation control.

viii. Earthing arrangements with earth resistance values.

ix. Reservoir Data.

a. Salient features

b. Type of Reservoir

1. Multipurpose

2. For Power

c. Operating Table with

1. Area capacity curves and

2. Unit capability at different net heads

(2)

Protection

i. Full description including settings for all relays and protection systems installed on the Generating Unit, Generator transformer, auxiliary transformer and electrical motor of major equipment included, but not limited to those listed, under Sec. 3 (General).

ii. Full description including settings for all relays installed on all outgoing feeders from Power Station switch-yard, tiebreakers, and incoming breakers.

iii. Full description of inter-tripping of breakers at the point or points of Connection with the Transmission System.

iv. Most Probable fault clearance time for electrical faults on the User's System.

(3)

Switch-Yard

i. Interconnecting transformers:

1. Rated MVA

2. Voltage Ratio

3. Vector Group

4. Positive sequence reactance for maximum, minimum and normal Tap.(% on MVA).

5. Positive sequence resistance for maximum, minimum and normal Tap.(% on MVA).

6. Zero sequence reactance (% on MVA)

7. Tap changer range (+% to -%) and steps.

8. Type of Tap changer (off/on load).

9. Neutral grounding details.

ii. Switchgear (including circuit breakers, Isolators on all circuits connected to the points of Connection).

1. Rated voltage (KV).

2. Type of Breaker (MOCB/ABCB/SF6).

3. Rated short circuit breaking current (KA) 3 phase.

4. Rated short circuit breaking current (KA) 1 phase.

5. Rated short circuit making current (KA) 3 phase.

6. Rated short circuit making current (KA) 1 phase.

7. Provisions of auto re-closing with details.

iii. Lightning Arresters

Technical data

iv. Communications

Details of Communications equipment installed at points of connections.

v. Basic Insulation Level (KV)

1. Bus bar.

2. Switchgear.

3. Transformer Bushings

4. Transformer windings.

(4)

Generating Units

i. Parameters of Generator

1. Rated terminal voltage (KV).

2. Rated MVA.
3. Rated MW
4. Speed (rpm) or number of poles.
5. Inertia constant H (MW sec./MVA).
6. Short circuit ratio.
7. Direct axis synchronous reactance X_d (% on MVA).
8. Direct axis (saturated) transient reactance (% on MVA) X'_d .
9. Direct axis (saturated) sub-transient reactance (% on MVA) X''_d .
10. Quadrature axis synchronous reactance (% on MVA) X_q .
11. Quadrature axis (saturated) transient reactance (% on MVA) X'_q .
12. Quadrature axis (saturated) sub-transient reactance (% on MVA) X''_q .
13. Direct axis transient open circuit time constant (sec) T'_{do} .
14. Direct axis sub-transient open circuit time constant (sec) T''_{do} .
15. Quadrature axis transient open circuit time content (sec) T'_{qo} .
16. Quadrature axis transient open circuit time constant (sec) T''_{qo} .
- 17.
- Stator Resistance (Ohm) R_a .
- 18.
- Stator leakage reactance (Ohm) X_1 .
- 19.
- Stator time constant (Sec).
- 20.
- Rated Field current (A).
- 21.
- Neutral grounding details.
- 22.
- Open Circuit saturation characteristics of the Generator for various terminal voltages giving the compounding current to achieve this.
- 23.
- Type of Turbine.
- 24.
- Operating Head (Metres)
- 25.
- Discharge with full gate opening (cumecs)
- 26.
- Speed Rise on total Load throw off(%).
- 27.
- MW and MVar Capability curve
- ii. Parameters of excitation control system:
- iii. Parameters of governor:
- iv. Operational parameter:
 1. Minimum notice required to Synchronise a Generating Unit from desynchronisation.
 2. Minimum time between Synchronising different Generating Units in a Power Station.
 3. Minimum block load requirements on Synchronising.
- (5)
- General Status
- i. Detailed Project Report.
- ii. Status Report.
 1. Topographical survey
 2. Geological survey
 3. Land

4. Environmental Clearance
5. Rehabilitation of displaced persons.
- iii. Techno-economic approval by Central Electricity Authority.
- iv. Approval of State Government/Government of India.
- v. Financial Tie-up.

(6)

Connection

- i. Reports of Studies for parallel operation with the State Transmission System.

1. Short Circuit studies
2. Stability Studies.
3. Load Flow Studies.

- ii. Proposed Connection with the State Transmission System.

1. Voltage
2. No. of circuits
3. Point of Connection.

(7)

Reservoir Data

- i. Dead Capacity
- ii. Live Capacity

B.2 Detailed System Data-Transmission

For STU/Transmission Licensees

(1)

General

- i. Single line diagram of the Transmission System down to 66KV.33KV bus at Grid Sub-station detailing:

1. Name of Sub-station.
2. Power Station connected.
3. Number and length of circuits.
4. Interconnecting transformers.
5. Sub-station bus layouts.
6. Power transformers.
7. Reactive compensation equipment.
- ii. Sub-station layout diagrams showing:
 1. Bus bar layouts.
 2. Electrical circuitry, lines, cables, transformers, switchgear etc.
 3. Phasing arrangements.
 4. Earthing arrangements.
 5. Switching facilities and interlocking arrangements.
 6. Operating voltages.
 7. Numbering and nomenclature:
 8. Transformers.
 9. Circuits.
 10. Circuit breakers.
 11. Isolating switches.

(2)

Line Parameters (for all circuits)

- i. Designation of Line.

1. Length of line (km).
2. Number of circuits.Per Circuit values.
3. Operating voltage (KV).
4. Positive Phase sequence reactance (pu on 100 MVA) X1
5. Positive Phase sequence resistance (pu on 100 MVA) R1

6. Positive Phase sequence susceptance (pu on 100 MVA) B1
7. Zero Phase sequence reactance (pu on 100 MVA) X0
8. Zero Phase sequence resistance (pu on 100 MVA) R0
9. Zero Phase sequence susceptance (pu on 100 MVA) B0

(3)

Transformer Parameters (For all transformers)

- i. Rated MVA
- ii. Voltage Ratio
- iii. Vector Group
- iv. Positive sequence reactance, maximum, minimum and normal (pu on 100 MVA) X1
- v. Positive sequence resistance, maximum, minimum and normal (pu on 100 MVA) R1
- vi. Zero sequence reactance (pu on 100 MVA).
- vii. Tap change range (+% to -%) and steps.
- viii. Details of Tap changer. (Off/On load).

(4)

Equipment Details (For all substations)

- i. Circuit Breakers
- ii. Isolating switches
- iii. Current Transformers
- iv. Potential Transformers /CVTs

(5)

Relaying and Metering

- i. Protection relays installed for all transformers and feeders along with their settings and level of co-ordination with other Users.
- ii. Metering Details.

(6)

System Studies

- i. Load Flow studies (Peak and lean load for maximum hydro and maximum thermal generation).
- ii. Transient stability studies for three-phase fault in critical lines.
- iii. Dynamic Stability Studies
- iv. Short circuit studies (three-phase and single phase to earth)
- v. Transmission and Distribution Losses in the Transmission System.

(7)

Demand Data (For all substations)

Demand Profile (Peak and lean load) for next 5 years.

(8)

Reactive Compensation Equipment

- i. Type of equipment (fixed or variable).
- ii. Capacities and/or Inductive rating or its operating range in MVar.
- iii. Details of control.
- iv. Point of Connection to the System.

B.3 Detailed Planning Data (Distribution)

For Distribution Licensees

(1)

General

- i. Distribution map (To scale). Showing all lines up to 11KV and substations belonging to the Supplier.
- ii. Single line diagram of Distribution System (showing distribution lines from points of Connection with the Transmission System, 132/11 kV sub stations 66/11 KV substations, 33/11 KV substations, and consumer bus in case of consumers fed directly from the Transmission System).
- iii. Numbering and nomenclature of lines and sub-stations (Identified with feeding Grid sub-stations of the Transmission and concerned 220/ 11 kV, 132/11 kV, 66/11 kV and 33/11 KV sub-station of Licensee).

(2)

Connection

i. Points of Connection (Furnish details of existing arrangement of Connection).

ii. Details of metering at points of Connection.

(3)

Loads

i. Details of major loads of 1 MW and above to be contracted for next 5 years.

ii. Demand profile of Distribution System (Current & Forecast)

Appendix C

Operational Planning Data

C.1 Outage Planning Data

C.1.1 Demand Estimates

For Distribution Licensees

Item

Due date/Time

(a)

Estimated aggregate month-wise annual sales of
Energy in Million Units and peak and lean demand in MW & MVAR
at each Connection point for the next financial year.

15th November of current year

(b)

Estimated aggregate day-wise monthly sales of
Energy in million Units and peak and lean demand in MW & MVAR
at each Connection point for the next month.

25th of current month

(c)

15 Minute block-wise demand estimates for the
day ahead.

9.00 Hours every day.

(2) Estimates of Load Shedding for Distribution

Licensee

Item

Due date/Time

(a)

Details of discrete load blocks that may be
shed to comply with instructions issued by SLDC when required,
from each connection point.

Soon after connection is made.

(3) Year ahead outage

programme (For the financial year)

(i) Generation outage programme for (SGS)

Item

Due date/Time

(a)

Identification of Generating Unit.

15th November each year

(b)

MW, Which will not be available as a result of
Outage.

(c)

Preferred start date and start-time or ranges

of start dates and start times and period of outage.

(d)

If outages are required to meet statutory requirement, then the latest - date by which outage must be taken.

(ii) Year ahead outage programme (Affecting Transmission System)

Item

Due date /Time

(a)

MW, which will not be available as a result of Outage from Imports through external connections.

1st November each year

(b)

Start date and start time and period of Outage.

(iii) Year ahead CPP's outage programme (Affecting Transmission System)

Item

Due date/Time

(a)

MW, which will not be available as a result of Outage from Imports through external connections.

30th November each year

(b)

Start date and start time and period of Outage.

(iv) Year ahead Distribution Licensees outage programme (Affecting Transmission System)

Item

Due date/Time

(a)

Loads in MW not available from any connection point. Identification of connection point.

15th November each year

(b)

Period of suspension of drawal with start date and start time.

(v) STU's Overall outage programme

Item

Due date/Time

(a)

Report on proposed outage programme to ERPC

15th February each year

(b)

Release of finally agreed outage plan

15th February each year

C-2. Generation Scheduling Data

For SGS

Item

Due date/ Time

(a)

Day ahead 15-minute block-wise MW/MVAr availability (00.00 - 24.00 Hours) of SGS.

9.00 hrs

(b)

Day ahead 15-minute block-wise MW import/export from CPP's.

9.00 hrs

(c)

Status of Generating Unit Excitation AVR in service (Yes/No).

9.00 hrs

(d)

Status of Generating Unit Speed Control System. Governor in service (Yes/No).

9.00 hrs

(e)

Spinning reserve capability (MW).

9.00 hrs

(f)

Backing down capability with/without oil support (MW).

9.00 hrs

(g)

Hydro reservoir levels and restrictions.

9.00 hrs

(h)

Generating Units 15-minute block-wise summation outputs (MW)

9.00 hrs

(i)

Day ahead 15-minute block-wise MW entitlements from Central Sector Generation Power Stations from ERLDC.

10.00 hrs

C-3 Capability Data

For SGS

Item

(a)

Generators and IPPs shall submit to STU up-to-date capability curves for all Generating Unit. On receipt of request from STU/SLDC.

(b)

CPPs shall submit to STU net return capability that shall be available for export /import from Transmission System

On receipt of request from STU/SLDC.

C-4 Response to Frequency Change

For SGS

Item

(a)

Primary Response in MW at different levels of loads ranging from minimum generation to registered capacity for frequency changes resulting in fully opening of governor valve. On receipt of request from STU/SLDC.

(b)

Secondary response in MW to frequency changes

On receipt of request from STU /SLDC.

C-5 Monitoring of Generation

For SGS

Item

(a) SGS shall provide 15-minute block-wise

generation summation to SLDC.

Real time basis

(b) CPPs shall provide 15-minute block-wise

export / import MW to SLDC.

Real time basis

(c) Logged readings of Generators to SLDC.

As required

(d) Detailed report of generating unit tripping

on monthly basis.

In the first week of the succeeding month

C-6 Essential and Non Essential Load Data

For SGS

Item

Due date/Time

(a)

Schedule of essential and non-essential loads

on each discrete load block for purposes of load shedding.

As soon as possible after connection

Appendix D:

Site Responsibility Schedule

Name of Power Station / Sub - Station:

Site Owner:

Site Manager:

Tel. Number:

Fax Number:

Item of Plant / Apparatus

Plant Owner

Safety responsibility

Control responsibility

Operation responsibility

Maintenance responsibility

Remarks

_____KV Switch-yard

All equipment including bus bars

feeders

Generating units

Appendix E:

Incident Reporting

First report

Date:_____

Time:_____

S.No.

Item

Details

1

Date and time of incident

2

Location of incident

3

Type of incident

4

System parameters before the incident (voltage, frequency, flows, generation etc.)

5

Relay indications received and performance of protection

6

Damage to equipment

7

Supplies interrupted and duration, if applicable

8

Amount of generation lost, if applicable

9

Possibility of alternate supply arrangement

10

Estimate of time to return to service

11

Cause of incident

12

Any other relevant information and remedial action taken

13

Recommendations for future improvement / repeat incident

14

Name of the organization

Related AI tags, queries and research notes

Translation